M-INSTR spec: per-instrument profile layer (InstrumentProfile) — rev 2.5.7 (2026-11-09)

Rev 2.5.7, 2026-11-09 (ET). PR-M6e. `ringBars` stays 2880 (`_RangeProfile` L31) and 1440 (`_ProfileStats` L47): bar counts, not derived. WEEKLY `RthSessions` on a profile whose sessions are filled and whose `rthSplit` is null is S7 (`sessions:no-rth-split`, action "use a day-session template for <root>"). ES/MES/NQ/MNQ on a 24-hour chart have `EquityIndexLegacy`, and a day-only chart has `ChartDaySession`, so they do not throw. `GlobexSessions` does not need a split; that is the existing source the user sets, not a new `[Input]` and not an automatic fallback. The bars that count as RTH stay on `isRthBar` (VPFUT: the equity-index value window, including holidays and early closes), so ES weekly results do not move. The KDE disagreement stays `> 4 × tick`. When profile stats are enabled, one Info line is `CFG PSTATS applied{disagree>4xtick ES-tuned default} undefined{} degraded{}`. Pins do not include these fields. History of the rev 2.5.6 header follows. **Rev 2.5.6, 2026-11-09 (ET). PR-M6d.** LEVEL and AVWAP no longer default `tickSize` to 0.25. The host already sets both from `tickSize()` (`startLevelStudy` L1477, `startAvwap` L2147), which is the Data1 profile tick (ES and MES 0.25). Gate 0 copies its own `tick` onto the embedded study in `Configure` (L471) before that study runs. `_AvwapManager` still rejects a tick that is not greater than 0 (L196). The level study does not throw: an unset tick is 0, which does not snap controls. Hosted ES and MES results do not move, and the pins do not include this field. History of the rev 2.5.5 header follows. **Rev 2.5.5, 2026-11-09 (ET). PR-M6c.** DAILY and REGIME read the profile's `sessionsPerYear`. Futures and stocks stay 252, so ES and MES are unchanged. Crypto uses the existing class constant 365 and spot FX the existing 260 (still marked VERIFY in §4.5; M6c does not pick a new number). The default lookbacks `{252,126,63}` are one year, half a year and a quarter, so they become `sessionsPerYear`, `sessionsPerYear/2` and `sessionsPerYear/4` (integer division) only while they are still that default. An explicit other length is left alone. `hvAnnualisation` moves off 252 the same way. REGIME's 63-session window stays; only the `sqrt(63/252)` denominator becomes `sqrt(63/sessionsPerYear)`. The fit z-scores that feature, so labels on a constant scale do not move; the raw feature's fit `zSd` does. Profile hashes already include `spy=`, and 252 is already the ES value, so the pins stay. `_TestGate0` and `_DvpEngineTest` do not use these fields. History of the rev 2.5.4 header follows. **Rev 2.5.4, 2026-11-09 (ET). PR-M6b.** Ettienne 08:04 ET, verbatim: "Use the template's listed session end converted to ET, so ES moves to 5:30 PM ET on that template; I accept the ES change." The FLOW tick-chunk boundary is that listed end in US Eastern. On the stale CME Globex template, 16:30 CT is 17:30 ET, so ES chunking moves 30 minutes from the old 17:00 ET. History of the rev 2.5.3 header follows. **Rev 2.5.3, 2026-11-09 (ET). PR-M6a.** Gate 0 no longer defaults `tick` to 0.25: 0 or unset writes an Error line, then throws. KD-02 passes the profile root instead of a hardcoded `"ES"`. The frozen 09:30–16:00 ET windows are unchanged and are refused with `sessions:window-not-covered` when the chart sessions, converted to ET, do not cover them. History of the rev 2.5.2 header follows. **Rev 2.5.2, 2026-11-08 (ET). Evidence only.** Ettienne 22:04 ET, verbatim: "We can accept todays runs as baseline". The Nov 8 native runs reported by Analyst are the documented MI-T15 baseline (revision log, §6.1, §10, §12.4). No code change. History of the rev 2.5.1 header follows. **Rev 2.5.1, 2026-11-06 (ET). APPROVED for implementation; M0–M4 merged. M5 own-contract controls approved by Ettienne 2026-11-06 12:13 ET (verbatim quote in the revision log and §12.4).** History of the rev 2.5 header follows. **Rev 2.5, 2026-11-04 (ET). APPROVED for implementation; M0–M4 merged.** Rev 2.5 re-bases the spec on merged `origin/main` @ **`d3c0c1a`** (merge of PR #98, PR-M4, final head `2418052465215c0795954aae03d4173b6959e20e`; first-parent merges #89 `6853398`, #90 `d2b41b7`, #91 `670e28d`, #92 `63cdff0`, #93 `ecc5436`, #95 `7ec8c95`, #96 `4500c67`, #94 `ed3e4a5`, #97 `5c9ccd2`, #98 `d3c0c1a`). Every cite marked **@d3c0c1a** was read on that commit; `_InstrumentProfile.Function.CS` is abbreviated **IP**, `_CustomSignal.Strategy.CS` **CS**, `tests/InstrumentTests/Program.cs` **IT** (InstrumentTests: 756 passed, 0 failed on the M4 head, .NET 8, box; full check: **23 suites, 4528 passed, 0 failed**). Where the code and this spec differ, the difference is flagged **DIVERGENCE** (§12.5). Older cites below are kept as history and say which commit they were read on. **A1 (Ettienne, 2026-11-04):** keep the strict version — stop any strategy on any bad stream; M4 enforces every refusal globally at StartCalc on every data stream for all `_CustomSignal`-based studies (§11.1 item 1, §12.4). **A2 (Ettienne, 2026-11-04):** `_TestPortMM.Signal.CS` deleted in M4; live-check-sheet Check 7 MM-side needs a replacement MM test signal before that check can run (§12.4). **A3 OPEN:** composite / feed exchange names (e.g. NYSEARCA, NASDAQGS, CMEGLOBEX, AMEX, BATS, CFE) are unmapped pending a live check of real `Bars.Info.Exchange` strings on dxFeed, plus the GMT chart VERIFY (§6.1, §12.4). Pins kept: 6E Default `55576df6a8a99c05`, ES Default `6cff053d1b215324`, AAPL `41886799d411dd0d`. History of the rev 2.4 header follows. **Rev 2.4, 2026-11-04 (ET). APPROVED for implementation; M0–M3 merged.** Rev 2.4 re-bases the spec on merged `origin/main` @ **`5c9ccd2`** (merge of PR #97, PR-M3; first-parent merges #89 `6853398`, #90 `d2b41b7`, #91 `670e28d`, #92 `63cdff0`, #93 `ecc5436`, #95 `7ec8c95`, #96 `4500c67`, #94 `ed3e4a5`, #97 `5c9ccd2`). Every cite marked **@5c9ccd2** was read on that commit; `_InstrumentProfile.Function.CS` is abbreviated **IP**, `_CustomSignal.Strategy.CS` **CS**, `tests/InstrumentTests/Program.cs` **IT** (InstrumentTests: 668 passed, 0 failed on `5c9ccd2`, .NET 8, box). Where the code and this spec differ, the difference is flagged **DIVERGENCE** (§12.5). Older cites below are kept as history and say which commit they were read on. Cost keying (§12.4 D1) was DECIDED by Ettienne on 2026-11-04: costs are keyed by asset class + root (`Future:ES`), implemented in PR-M4. **D2** (profileHash over sessions in exchange time) was DECIDED by Ettienne on 2026-11-04, implemented in PR-M4 (§6.1). **MC verification 2026-11-04 (Ettienne, `5c9ccd2`):** MI-T1 PLEditor compile passed, `eMTPA_SF_Root` / `eMTPA_MCSC_USD` no longer VERIFY; the 6E MC run reproduced its profileHash byte-for-byte on .NET 8; the Portfolio Trader per-symbol test passed (§12.3). History of the rev 2.3 header follows. **Rev 2.3, 2026-11-03 (ET). APPROVED for implementation: all questions decided (Q1–Q10, N1–N6). Rev 2.3 is final and ships with PR-M1b** (#90 commits rev 2.2 byte-identical). Rev 2.2 was based on rev 2.1 as merged in `origin/main` @ `6853398` (PR #89, PR-M0; #88 merged as `cd7e50a`) and adds the PR-M0 dump results (§4.4.2). Data inputs still pending from Ettienne: Q3 IBKR per-side fees for the non-ES/MES roots and the Q7 bid/ask start dates; they do not block PR-M0 to PR-M3. Checked read-only against `github.com/ettiennehugo/MultiCharts.Net` `origin/main` @ `7dbdb1b` (merge of PR #85). PR #88 ("Land #86 + #87 on main") was OPEN at rev 2.1 and has since merged (`cd7e50a`); its code is quoted where marked "#88 head", and `_MtfMath.SymbolRoot` was at L108–119 (doc L108, method L109–119) on `6853398`, i.e. **main before #90**; after #90 (PR-M1a, head `a265f4d`) it is doc L108–119, method **L120–135**. Every file:line below was read on `7dbdb1b` unless marked **VERIFY** or "#88 head". **Prerequisite: PR #88 merges before any M-PR.** #88 changes `_MtfMath` (around `SymbolRoot`, L29–39), `_CustomSignal` L1731–1830 (incl. the same-root check L1776–1785), `_DvpEngine`, the DVP golden fixtures and configHash. After it merges, `_CustomSignal` line numbers from about L1740 shift by up to about 50 lines; each M-PR re-reads its cites on the main it targets. Every M-PR targets `main` and is opened only after the previous one merged (no stacked PRs, §11).

Sources: the repo (every `*.CS`, `specs/*.md`, `tests/*`); `specs/instrument-profile-spec.md` rev 3 (2026-10-24, rev-3 Q1–Q11 DECIDED); `specs/m-dvp-spec.md` rev 6; `specs/m-port-spec.md` rev 2.7; the MultiCharts .NET Programming Guide, "Instrument properties" list (just before §4.2.4 StatusLine; box copy, not in the repo); Ettienne's typed API dump output `_McApiDump3.output.txt` (@ES, 2026-11-01; **box copy only, not in the repo**, and produced by a reflection dump, so it is cited as Ettienne's run output, not as a repo file; PR-M0 commits typed dump output as a fixture); CME contract-spec pages read 2026-11-02 (§8); the box research note `/workspace/m-instr/mc-sessions-research.md` (MC sessions: Programming Guide v1.1 §4.2.2–4.2.3, §4.7.4, MC wiki and forum staff answers, MC-1439; 2026-11-02); Ettienne's rulings (ES $17 and MES $2.50 per round trip, 2026-11-01; sessions from the chart, 2026-11-02; Q3–Q10 accepted, 2026-11-02).

Why a new file. Rev 3 treats "Future = today's ES behaviour" as an identity (IP-4.4) and classifies on `Category.ToString()` substrings (IP-1.1, IP-2.2). The 2026-11-01 audit found the futures path itself is ES-only (CME equity-index session clock, quarterly H/M/U/Z roll, third-Friday expiry, the root parser, the clusters). This spec **supersedes rev 3 where they conflict (§1.2)**; rev 3's DECIDED answers stay in force otherwise. Nothing from rev 3 I1–I6 is coded (`rg 'EInstrumentClass|InstrumentProfile' -g '*.CS'` finds nothing); rev-3 PR-I0 `_InstrProbe` is on main and is retired by PR-M1e.

Scope. M-INSTR is a **base service** in `_CustomSignal`. At `StartCalc()` (and, for session data, on the first `CalcBar` of each stream, §4.5) it: 1. reads every used stream's instrument through the **typed** API (`Bars.Info` / `BarsOfData(n).Info`, `IInstrumentSettings`) and the stream's **chart sessions** (`Bars.Sessions`, `IROList<SessionObject>`); 2. builds one immutable `InstrumentProfile` per stream from detection, the chart's sessions, a per-root **exchange-facts** preset (no times) and the user's **cost** figures; 3. logs detection and the applied configuration; 4. hands each engine its effective settings; 5. throws a clear `EngineNotSupportedException` where an engine cannot apply (one rule, R13).

Tuned parameters (thresholds, bandwidths, gap sizes, degrees) are **not** in profiles (R5).

Not touched. The PriceAction library. No new `[Input]`s. M-PORT behaviour (§9). The `_TestGate0` canary and the DVP D9a summary stay the documented ES baseline (rev 2.5.2, MI-T15). The M6b order-flow chunk boundary on the stale Globex template is the exception already in §7. The older post-#88 exports were not found. ES fixtures in `tests/*` are unchanged.

Revision log

Rev	Date (ET)	Change
1	2026-11-02	First draft (62-row audit, rev-3 conflicts C1–C8, Q1–Q10 OPEN).
2	2026-11-02	Coder review (37 items, Appendix A) applied, each re-verified on `7dbdb1b`. **Sessions come from the chart (Q1/Q2 DECIDED by Ettienne):** no session/RTH windows in any preset; trading day, day open and session end derive from typed `Bars.Sessions`; RTH-window presets dropped. Q3–Q10 DECIDED (accepted recommendations). One throw-vs-degrade rule (R13). Lint scoped to MC enums with an allow-list. `_InstrProbe` retired in its own PR. Gate0Tests updated with the G0 tick change. ringBars kept literal. §9 aligned with the `willArm` code; cost floor is new opt-in behaviour. Audit 70 rows (8 added, 5 corrected, counts fixed). F1 widened, F3 second bug. `pointValue` → `bigPointValue`. M1 split four ways (+ probe retirement), M6 per engine. New questions N1–N6.
2.1	2026-11-03	Ettienne 2026-11-03 09:47 ET: "1 to 5 is fine, on 6 keep that dump in the same directory as the created standard log file". N1–N5 DECIDED as recommended; N6 DECIDED: `_InstrDump` writes `MCInstrDump_<symbol>_<yyyyMMdd-HHmmss>-<pid>.txt` into the directory of the host's standard `MCSignal*.csv` log, through the host's existing resolver (§4.4); no new hardcoded path. N4 becomes approved degraded mode D7. Status APPROVED for implementation.
2.2	2026-11-03	Base: rev 2.1 as merged in `origin/main` @ `6853398` (PR #89). **§4.3 root rule rewritten** (rev 2.1 steps 3–4 turned ESZ2024 into ESZ2024): leading digits + letter run, drop a final month letter followed by a digit when the run has ≥ 2 letters; MI-T2 gains ESZ2024, 'ES 12-24', ESZ24:XCME, CLZ2026, EUR/USD, BRK.B, Z4. **PR-M0 dump results** (Ettienne's 13 `_InstrDump` runs, §4.4.2): CONFIRMED `Bars.Sessions` filled at StartCalc and identical on the first bar (R12 now reads at StartCalc, re-checks on the first bar), session/bar times follow the chart time zone, per-series sessions, multiple sessions per day kept, `SessionName` value, 6E priceScale within the 1e-9 tolerance, SPY facts, CL/GC tick and big point value. **New MC behaviour:** on the GLOBEX template `LastBarInSession` did not fire at the 15:15 break, only at 16:00 → new **R14** (session ends from the Sessions list end times) and **MI-T16**. Still OPEN (awaiting Ettienne): Portfolio Trader second symbol not dumped; 6E Default session vs session-end bars (QuoteManager zone). B1/B2 not run (commented out in the build).
2.5.7	2026-11-09	**PR-M6e.** ringBars literals kept (2880, 1440). `InstrumentSessions.RequireWeeklyRth` throws S7 only when WEEKLY source is `RthSessions` and the stream's profile has sessions but no `rthSplit`. Profiles off, sessions not yet filled, and `GlobexSessions` do not throw. `isRthBar` is unchanged. PSTATS logs the 4-tick ES-tuned note once at Info when enabled; the comparison stays `4 * tick`. No new `[Input]`, no fee or pin change. Canary `b04cc95a` and D9a `7fbe5bff` do not enable weekly profiles or profile stats.
2.5.6	2026-11-09	**PR-M6d.** `_LevelStudy.tickSize` (L80) and `_AvwapManager.tickSize` (L154) have no initializer. Hosts keep assigning `tickSize()` (CS L1477, L2147). Gate 0 keeps `study.tickSize = tick` (L471). Avwap `Configure` still returns `AvwapManager tickSize=…` when the tick is not > 0. The level study does not gain a new refusal. No new `[Input]`, no fee or pin change. ES/MES hosted results stay on the profile tick, so canary `b04cc95a` and D9a `7fbe5bff` are unchanged.
2.5.5	2026-11-09	**PR-M6c.** DAILY `ApplySessionsPerYear` and REGIME `sessionsPerYear` read `InstrumentProfile.sessionsPerYear` (already set in `Build` from the class constant: 252 futures/stocks, 365 crypto, 260 spot FX **VERIFY**). Host `applyYearLength` runs from `startDaily` and `startRegime` after `buildProfiles`. Default `{252,126,63}` becomes `{spy, spy/2, spy/4}` and `hvAnnualisation` 252 becomes `spy`, only while those fields are still the defaults. The 63-session REGIME window is not scaled. ES/MES stay 252, so canary `b04cc95a` and D9a `7fbe5bff` are unchanged. No new `[Input]`, no fee or pin change.
2.5.4	2026-11-09	**Ettienne 08:04 ET, verbatim:** "Use the template's listed session end converted to ET, so ES moves to 5:30 PM ET on that template; I accept the ES change." **PR-M6b.** `TickChunkBoundaryEt` is no longer the ES boundary. The host sets it from the Data1 template's listed session end (the latest weekday session `End` after conversion from the bar zone to US Eastern). Stale CME Globex (listed end 16:30 CT) → **17:30 ET**. A template whose listed end is 16:00 CT stays 17:00 ET. 6E uses its own listed end the same way. Spot FX throws (Error line first) and gets no boundary. The old §7 / §11 claim that ES stays identical at 17:00 ET is withdrawn for this template. Profile hashes, the roll digest, ES floor 14 and MES floor 20 do not include the chunk boundary, so those pins stay. Order-flow chunk windows on that template are the golden that changes: Sunday 17:00→Monday 17:00 ET becomes Sunday 17:30→Monday 17:30 ET. `_TestGate0` and `_DvpEngineTest` leave order flow off, so their canary and D9a numbers are not this golden.
2.5.3	2026-11-09	**PR-M6a.** `_Gate0 tick` has no default (`L358`); `Configure` writes the Error sink, then throws `G0: tick must be > 0` when tick is 0, unset, or not finite. The host sets `tick` from `tickSize()` and `root` from the Data1 profile (`p.root`; symbol root only when profiles are off). KD-02 (`L1214`) passes that root into `_DvpEngine`. S3: `InstrumentSessions.RequireGate0Rth` after the sessions are converted to ET; reason `sessions:window-not-covered`; action "use a session template that covers 09:30–16:00 ET". The window expressions stay (`IsRthTime` L484, 30-minute periods L624, flat 15:55 L1047). No new `[Input]`, no fee or pin change. ES result on a covering template is unchanged. Native MI-T15: no trade-list SHA exists for baseline `b04cc95a` (owner skipped the export); a rerun of that same canary is expected to reproduce 35,109 trades and net PF 0.84 because the ES tick is unchanged.
2.5.2	2026-11-08	**Ettienne 22:04 ET, verbatim:** "We can accept todays runs as baseline". Evidence only. The post-#88 G0/D9a exports were missing, so these Nov 8 native runs, as reported by Analyst from the MiniPC at frozen `388314a`, are the documented baseline in their place. Not box-reproduced. **Compile All:** PLE Recompile All 14:15–14:16 ET, "Build succeeded", no errors; new assembly `CompAssms\gdbhzsw4\gdbhzsw4_CS.dll`, 2,103,296 bytes; HEAD `388314a` clean; `PLStudies.csproj` SHA-256 prefix `ec02edfb` unchanged. **G0:** `_TestGate0` Canary, @ES 1-minute, run `b04cc95a`, 2016-11-17 through 2024-12-31 (dxFeed data start; the later start is accepted), 35,109 trades, Strategy Performance Report net profit factor 0.84 (at most 1.0, plumbing pass), commission $2.25 per side, slippage $6.25 (0.5 tick) per side, Bar Magnifier 1 tick. The variant number was not reported, so this is that one canary run and not the IP-T46 set. The owner did not choose a trade-list export, so that hash is skipped. No trade-list SHA-256 exists for baseline `b04cc95a`. A future identity comparison needs a fresh export. The documented canary baseline is the reported performance numbers only. **D9a:** `_DvpEngineTest`, Data1 @ES 15-minute and Data2 @ES 60-minute, CME Globex template, Trade Volume, requested 2016-01-01 through 2024-12-31, Max bars back user-defined 50, Journal. Runs `7fbe5bff` and `86353b4f`: setup `deviation(data1=0 data2=0 chartStart=1 maxBarsBack=0 root=0)`; `chartStart=1` is only the dxFeed start (`firstBarEt` 2016-11-21 22:00), accepted. RUN-END 2024-12-31 16:00, bar 188581. Zero Error lines. `configHash 80cf8ab6ad9ba5cd` (`streams` 1 and 2; not the synthetic single-stream `6221dc6538c1eedb`). `profileHash 020deca78d6392b7` on both streams. Data1 `matchPooled` 95/95, `warmEnd` 2017-01-31. Data2 87/87, `warmEnd` 2017-10-01. After stripping `runId`: DEGREE 196/196, D9A 199/199, CAL 62/62 identical. The only diffs are `elapsedMs` and a run-1-only one-time "VP hook Data1 not connected" warning. Run `0ea50cd4` (Max bars back resolved 5049) is not the baseline. `BaselineDvpOff` timing was not rerun (already decided, runs `4247ff31` / `5f06042a`). **Open.** MC-internal pre-execute volume-profile snapshot sync at the realtime transition. No user-study frame in the stack. Not reproduced in the historical-capped baseline runs. Ettienne, 2026-11-08 about 22:06 ET (answer t515), verbatim: "The exception only blocks live-readiness, not the merge". If it recurs or stays unresolved it blocks live-readiness. It is not an M5 merge gate. Reported from MultiCharts' own traces (StudyRunner / Charting / MessageCenter, MC .NET64 16.0.29867.400), 18:01:05.349 ET, study `_TestGate0 (@ES-1 Minute)`, PID 9640, one occurrence: `System.ArgumentNullException: Value cannot be null. Parameter name: source`, at `System.Linq.Enumerable.Select`, `ChartingPlugin.Snapshot.Profile..ctor(IProfile)`, `VolumeProfilesSnapshots._full_sync`, `Syncronize`, `PriceSeriesImpl.OnBeforeRTCalc`, `PreExecuteImpl`, `SignalObject.PreExecuteImpl`, `DoExecute`, `CUserStudyImpl.Body`. Then `SwitchOff in _TestGate0` at 18:01:06 on `dxFeed:@ES 1 Minute`. Analyst's inference, labeled as inference: the run was `eee518cc`, the only run whose data reached 2026-11-06; the 2024-capped runs `b04cc95a`, `7fbe5bff`, and `86353b4f` do not show it. Check 7 remains separate and unrun. The ES 9-session pin `020deca78d6392b7` is what this Globex-template chart printed; see §6.1.
2.5.1	2026-11-06	**Ettienne 12:13 ET, verbatim:** "I approve separate expiry/roll controls in M5, preserving other calendar events, explicitly marking unsupported contract-expiry status unavailable, and rejecting explicit unsupported expiry/roll queries." Own-contract QuarterlyExpiry/RollWeek (`EventConfig.ownContract`, default on) are separate from shared calendar rows (MonthlyOpex, VixExpiry, IndexRebalance, HalfDay, RussellRecon; `EventConfig.sharedCalendar`, default on). Turning one off does not drop the other. No inferred per-asset-class event mask and no new `[Input]`. A profile with enforcement on and no roll facts does

Rev	Date (ET)	Change
		not inherit equity-index expiry dates. `Step/Classify` keep running and set `EOwnContractStatus.Unavailable` (`isRollWeek` / `isQuarterlyExpiry` are null, not false). An explicit own-expiry query (`ContractRollDate`, `ContractExpiryDate`, `IsRollWeek`, `SessionsToExpiry`, `InSpan(RollWeek)`) logs Error then throws `unknown-field:roll`. `SessionsToExpiry` on `NyseStandIn` still refuses `calendar` (S4) and does not count. ES/MES/NQ/MNQ and `InstrumentConfig.enabled == false` stay on the equity-index quarter. `EventConfig.Canon` gains `shared` and `own` (default 1/1). TableSha, profileHash pins, and the DVP golden configHash are unchanged.
2.5	2026-11-04	Re-based on merged main `d3c0c1a` (PR #98, PR-M4, final head `2418052`). **M4 DONE:** every §6.3 refusal is Error + throw globally at StartCalc on every data stream for all `_CustomSignal`-based studies (**A1**, Ettienne: "Keep the strict version: stop any strategy on any bad stream"); the `INSTR ds(n) log-only…` line and the null-profile / `profileHash(…=none)` path are gone. Refused: Unsupported/unmapped class; invalid `InstrumentConfig` (incl. old root-only cost keys → `unknown-field:config`); empty root; `tick` (PriceScale ≤ 0 / tick ≤ 0, no preset); non-USD (`currency`); unresolved bar/exchange zone (`tz:bar` / `tz:exchange`); missing or changed sessions; failed typed/session reads; `mismatch:*`; `calendar` / `months` when rules need them. Still runs: futures with no preset (cost/floor Unknown); USD stocks on NYSE/NASDAQ/ARCA. **`exchangeZoneId0f`** (CS L2587–2615): split the exchange name on non-alphanumerics, exact case-insensitive token match, first matching token wins; Chicago CME/XCME/GLOBEX/CBOT/ECBOT/CBT/XCBT/CBOE; New York NYMEX/COMEX/NYSE/NASDAQ/ARCA; anything else → `tz:exchange`. Composite names (NYSEARCA, NASDAQGS, CMEGLOBEX, AMEX, BATS, CFE) unmapped pending live dxFeed strings (**A3 OPEN**) + GMT chart VERIFY. **`InstrumentStreams.Start:`** sets `m_started`/`m_pending` only after `Guard(Build)` returns (a refused stream is never marked started). **DST tests** (summer, spring-forward gap, fall-back ambiguous read as standard) keep 6E pin `55576df6a8a99c05`. Pins confirmed in IT: 6E `55576df6a8a99c05`, ES `6cff053d1b215324`, AAPL `41886799d411dd0d`. **A2:** `_TestPortMM.Signal.CS` deleted at Ettienne's request; MI-T14 allow-list entry removed; live-check-sheet Check 7 MM-side needs a replacement MM test signal (open). `_RootDump.Strategy.CS` is on main (`e1f5b11`, when `_InstrDump` was retired); its four rows stay the MI-T2/MI-T3 fixtures. D1/D2 shipped. Full check: **23 suites, 4528/0**.
2.4	2026-11-04	Re-based on merged main `5c9ccd2` (M0, M1a–e, M2 #96, M3 #97 and #94 merged). **M2 (Ettienne's decisions, #96):** 6E roll = `BusinessDaysBeforeExpiry(5)` (§8.1, replaces "Mon of the expiry week"); CL/GC/ZN/6E use `ECalendar.NyseStandIn`, logged on INSTR as `calendar: NYSE stand-in (CME calendar pending M8)` until PR-M8; a preset with expiry/roll rules and no calendar refuses `calendar` at Build, no weekday fallback anywhere (R6); `expiry-worked.csv` roll column = preset rules, not vendor dates (MI-T4). **M3 (#97):** reason code `months` (§6.3); RUN suffix `profileHash(ds=h;…;ds=none)` outside configHash (§6.1); log-only line format; enforced set = `mismatch:*` + `sessions:changed`, everything else a log-only Warning with a null profile (§6.2, two-phase Build); INSTR example gains `root(<r> rootSrc=…)` and `dataNotes(…)`, ES and 6E lines pinned as the tests pin them; profile numbers printed/hashed with `InstrumentText.Fact` (G15), `Num` ("R") for diagnostics only; the profile carries the preset's tick/bpv after the cross-check; §4.3 6E tick text corrected (profile `5E-05`; raw .NET Framework "R" `5.0000000000000009E-05`); pinned profileHash ES `020deca78d6392b7`, 6E `8e8b3e9b0e230e12`; cost floor only with a matched preset, else `floor(Unknown)` (-1), and a new M4 requirement to log why; costs keyed by root only (**OPEN decision D1**, §12.4); Portfolio Trader per-symbol Sessions/ASymbolInfo2 test before M4; `eMTPA_SF_Root` **VERIFY** (not in the PG or any dump) until MI-T1/MI-T15; tz labels `unresolved` / `unavailable(<ExceptionType>)`; ClassOf stays in `_CustomSignal`, unmapped category is a log-only Warning in M3 and must be enforced in M4; `MonthMask.Of` throws without an Error line on static preset data only (accepted). §11 statuses with PR numbers; #94 listed as an extra; PR-M4 scope made explicit (FALLBACK_TICK sites, `tickFromMinMove` PriceScale==0 path, floor-Unknown reason, ClassOf enforcement, log-only → enforced, PT test, `eMTPA_SF_Root`). New §12.4 (open decisions), §12.5 (code-vs-spec divergences). Standing rules restated after R14. **D1 DECIDED (Ettienne 2026-11-04, same revision):** costs keyed by asset class + root, key `<Class>:<ROOT>` (defaults `Future:ES` 1700, `Future:MES` 250); a row whose class does not match the symbol is not used (`cost(Unknown)`, `floor(Unknown src=cost-class-mismatch)`, one Warning); root-only keying is M3 behaviour; firm PR-M4 scope (§5.3, §9, §11.1 item 8, §12.4); configHash unaffected (no configHash includes `InstrumentConfig.Canon()` on `5c9ccd2`); profileHash unchanged for ES/MES/6E, changes only where a cross-class cost was shown (e.g. a stock `ES`). **D2 DECIDED (Ettienne 2026-11-04):** profileHash hashes the sessions (and a `ChartDaySession` rth) converted to the **exchange** time zone (`exchangeTimeZoneId(Info.Exchange)`, strict lookup), so one instrument hashes the same on any chart/PT zone; new refusals `tz:bar` / `tz:exchange`, never a silent fallback; INSTR keeps bar-time `sessions(…)` and adds `xsessions(…)`; expected pins after M4: 6E Default `55576df6a8a99c05` on any zone, ES Default `6cff053d1b215324`; IT pins with exchange-time input unchanged, new tz-invariance tests (§6.1, §11.1 item 9, §12.4, V15). **MC verification (Ettienne 2026-11-04, `5c9ccd2`):** MI-T1 Compile All passed (after the one-time import of `_InstrumentProfile.Function.CS`); `eMTPA_SF_Root` and `eMTPA_MCSC_USD` compile, VERIFY dropped; `rootSrc=mc` on 6E, ES, AAPL; MI-T15 6E chart hash `55576df6a8a99c05` reproduced on .NET 8 (runtime-stable; `8e8b3e9b0e230e12` is the synthetic 9-session list); PT per-symbol test passed (@6E, @ES, AAPL; ES `e1927ccf1ea6f41e`, AAPL `41886799d411dd0d`), PT open item closed; PORT publish lines stamped `0001-01-01` / zone `?` (cosmetic, M4 item 10).
2.3	2026-11-03	Prepared after #90 (which commits rev 2.2 byte-identical); **final, ships with PR-M1b**. Header: the `SymbolRoot` L108–119 cite is main **before #90** (`6853398`); post-#90 location L120–135. §4.3/MI-T2: `MESH26` example replaced by `MESZ26`/`MESH27` → `MES`, as in the #90 tests. **@6E zone RESOLVED:** QuoteManager custom sessions with ET hours entered as CT (user config error, not MC behaviour); Ettienne switched to exchange sessions and will rerun; the 6E fixture is marked superseded. Folded in Coder's `mc-session-templates-research.md`: the built-in E-mini GLOBEX template is stale (2021 halt removal, pre-2012 15:30–16:30 row) and sits on the D9a/canary chart (environment caveat, engines take sessions as given); "Combined" = two sessions per day; templates editable in QuoteManager → Tools → Session Templates; `LastBarInSession` fires on rows with Session End ticked (inferred, not documented) as R14's rationale, with sources. Portfolio Trader @CL stays OPEN. **MC stores the root (credit: Ettienne):** `Bars.Info.ASymbolInfo2` (`MTPA_MCSymbolInfo2`, PG §4.2.3, pg11.txt L901, L3828–3867) becomes the M2 resolver's primary root, with the #90 parser as fallback when empty or flagged in `UnsupportedFields` (typed); `!SameRoot` → Warning, MC wins. §4.4 B1/B2 lead with `SymbolExpiry`/`SymbolCurrency`; new B3 cross-checks `SymbolCategory`/`MinMove`/`PriceScale`/`BigPointValue` vs `Bars.Info`/`IInstrument`. **Resolved by Ettienne `_RootDump` 2026-11-03, relayed by Coder (§4.4.3):** SymbolRoot/Currency/Category/Exchange/MinMove/PriceScale/BigPointValue CONFIRMED; parser = pure fallback on `(UnsupportedFields & eMTPA_SF_Root) != 0` or empty root; SymbolExpiry placeholder rule (B1); currency from SymbolCurrency (B2); MC-internal check PriceScale × Info.PriceScale = 1 ±1e-9 and BPV equal, else throw; MI-T2/MI-T3 fixtures; removed from §12.3. **@6E CLOSED:** rerun `01M6HTKV3SV1Q5QDXBFTCH8D3K.txt` (17:00–16:00 Exchange, ends 16:00, matches ES; `bvo(TickCount)` so instrument/session facts only) is the 6E fixture; `01M6HSDCCJZXJHNB165KSF6Z5B.txt` discarded; 6E rerun removed from §12.3. Setup note: restart MC after changing QuoteManager sessions (cached).

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0. Rules (normative; every PR must comply)

- **R1 Typed API only.** Never use MultiCharts enums or API members as strings: no `Enum.Parse`/`TryParse`/`GetNames` on an MC enum, no `<MC enum>.ToString()` used in a comparison, parse or hash, no reflection by member name on MC types. Formatting an MC enum into a human-readable log line is allowed (nothing reads it back). Repo-own enums parsed from protected config are outside R1 (§10 MI-T14 allow-list). Where a typed member is uncertain it is **VERIFY** with a typed dump line (§4.4); a line that does not compile is the answer (the member does not exist). Never guess.
- **R2 Build hygiene.** New `.CS` → `<Compile Include>` in `PLStudies.csproj` + one-time PowerLanguage Editor import (listed in the PR). A research draft that remains in `sandbox/` follows `docs/research-strategy-header.md` and stays out of that list until the owner approves promotion. New test projects → `PLStudies.Tests.sln`, `CheckForOverflowUnderflow=true`, real-source linking via a `gen.sh` awk region (`// <instr-core>` … `// </instr-core>`), as in `tests/InstrProbeTests`. C# 7.3.
- **R3 PriceAction unchanged.** The profile does not touch `EPriceActionPreset` or the PriceAction library; VPFUT keeps its existing preset call.
- **R4 No new `[Input]`s.** Settings go through protected config objects (`InstrumentConfig`, §5.3), like `EventConfig`, `RegimeConfig`, `DailyConfig`, `DvpConfig`, `PortMmConfig`.
- **R5 Self-calibration over tables.** Presets hold **exchange facts only**: tick, big point value, currency, contract months, expiry rule, roll rule, holiday calendar (where verified), cluster. **No session or RTH times** (Q1/Q2). ES-tuned values (e.g. `RollGapTicks` 8, `KdeBandwidthTicks` 6, `_ProfileStats` 4-tick disagreement) stay as today with a CFG note "ES-tuned default", or become relative to a running statistic in a later PR (§7).

- **R6 Auto-detect, apply, log, refuse.** No silent fallback. `TICK_FALLBACK = 0.01` (`_CustomSignal` L29, `resolveTick` L2287–2301 with a bare `catch {}`) becomes an Error + throw (PR-M1b; **done** in #91: CS `resolveTick` L2347–2368 @5c9ccd2, Error line via `traceAlways` then throw). **No weekday fallback anywhere (rev 2.4, #96):** every expiry/roll computation needs an assigned business-day calendar. `InstrumentCalendars.Require` (IP L450–455) and `InstrumentRules.requireCalendar` (IP L467–471) refuse with reason `calendar`; Build calls `Require` whenever the profile has an expiry or roll rule (IP L803), so a preset (or override row) with roll rules but no calendar (`Observed/None`) is refused at profile build. The old weekday default gave wrong dates (CLF26 12-22, ZNH24 03-20, ZNZ25 12-22; IT L209, L228). In M3 that `calendar` refusal was log-only (§6.2); **M4 enforces it** (A1). Known fallbacks audited and **removed in PR-M4** (§11): `FALLBACK_TICK = 0.01` in `_VolumeProfileBase` / `_VpConfluenceEngine`, and CS `tickFromMinMove` returning `MinMove` when `PriceScale == 0` (and `Math.Abs` of a negative ratio).
- **R7 Volume basis.** Futures (`Future`/`FutureRolover`) on Build Volume On = Tick Count → Error + throw; non-futures on Tick Count are tagged `TickCount`. **Already on main** (`_VolumeProfileBase` L1429–1461, L1505–1506); M-INSTR only moves it into `volumeBasis`.
- **R8 Sizing.** 1% risk per trade; hard max-qty/max-notional caps reduce size (`sizeForRisk`, L2921–2953). The portfolio path (`_PortMM`, deny-not-reduce) is unchanged. Details and the actual code path: §9.
- **R9 Costs.** ES $17.00 and MES $2.50 per round trip (Ettienne). Other roots: method DECIDED (Q3), figures need Ettienne's IBKR per-side fees per root (data input).
- **R10 Logs.** One `File.AppendAllText` per line; skipped entries at **Info**; doubles formatted with `CultureInfo.InvariantCulture`.
- **R11 Facts.** Cite file:line; exchange facts carry a source or **VERIFY**.
- **R12 Sessions come from the chart (DECIDED, Q1/Q2).** Session times, the trading-day boundary and the session end come from the stream's chart sessions via typed `Bars.Sessions` / `SessionObject`. The chart's session template is the source of truth (normally the instrument's MC "Default"). Engines do not second-guess it and do not warn about which template the user chose. Sessions are per stream (each data series has its own, **CONFIRMED** §4.4.2; in Portfolio Trader each symbol has its own, with one global time zone) and are read **in StartCalc and re-checked on the first CalcBar** of each stream (both reads identical in all 13 dump runs, **CONFIRMED** §4.4.2; a difference throws `sessions:changed`, S1). `Bars.Sessions` exists (it compiles on main) but is documented only on the MC forum; its behaviour is now taken from the PR-M0 dump (§4.4.2), not from the documentation. The template name (`Info.SessionName`, Programming Guide) is logged and never parsed. Holidays cannot be read from code, so holiday calendars stay in the repo as calendars only (PR-M8). The DV/MTF session-mismatch Warning between streams (#86/#88 `checkHtfSessions`), including its MC-1439 / Realtime-History Matching caveat, stays.
- **R13 Throw vs degrade (one rule).** An engine **throws** when it cannot apply to the instrument or to the chart's sessions. It runs **Degraded** only in a mode Ettienne has approved, which still produces correct results from reduced inputs; each such mode is listed in §6.4 with what is reduced. An output that is undefined for the given sessions (e.g. an overnight high on a chart with no overnight segment) is NaN, logged once at Info, and is not a degrade; §6.4 lists every such output.
- **R14 Session ends come from the Sessions list (NEW, rev 2.2).** A session ends at the `EndTime` of a `SessionObject` in the stream's Sessions list, not where `Bars.LastBarInSession` happens to be true. Observed (§4.4.2, `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`): on the CME GLOBEX template (sessions 17:00–15:15 and 15:30–16:30 CT) `LastBarInSession` was true only on the 16:00 bar, **not at the 15:15 break**. The session reader and every session-reset engine (VPFUT, AVWAP, FLOW, TOD, RANGE, WEEKLY, DAILY, DVP, G0, REGGATE, ROLL keying) therefore assign each bar to the session containing its close (start exclusive, end inclusive) and detect a session end as: (a) a bar whose close equals the session's `EndTime`, or (b) the first bar that falls in a different session than the previous bar (the end is then attributed to the previous bar), or (c) `LastBarInSession` true (kept only as an extra early signal for the last bar when data stops before the listed end, e.g. ES data ends 16:00 inside a session listed to 16:30). `LastBarInSession` alone is never the only signal. A reset fires **once per (stream, trading day, session index)**: the two-series run logged the same 2026-10-26 16:00 Data1 bar as a session end twice (`01M6HSDBFH2D6H21PMYXYKFJHV.txt`, CalcBar called once per stream update), so resets must be idempotent. Tested by MI-T16. **Rationale (rev 2.3):** each template row has a **Session End** checkbox and MC staff say to tick it on the row that ends the trading day (forum t=45178, https://www.multicharts.com/discussion/viewtopic.php?t=45178); ticking it on every row gave a break at every row (t=5890, https://www.multicharts.com/discussion/viewtopic.php?t=5890). `LastBarInSession` therefore most likely fires only at rows with Session End ticked, which matches the GLOBEX observation (15:15 row presumably unticked). This is **inferred, not documented** (research `/workspace/m-instr/mc-session-templates-research.md` §6), and the tick is not exposed by `SessionObject` (§4.1), so the list `EndTime`s are the only readable boundary; live, the flag can also be late or missed when no tick prints in the last bar (staff, https://www.multicharts.com/discussion/viewtopic.php?t=50358; https://www.multicharts.com/discussion/viewtopic.php?t=48918).

Standing rules (restated, rev 2.4; every PR still checks them): - **No MC enums or API values as strings** (R1): typed members only; MC enum labels are formatted by the host for log lines and never read back (IP header L8–11; lint MI-T14). - **Sessions come from MC session data** (R12): the chart's template (normally Default) via typed `Bars.Sessions`. The only code-side windows are the accepted ES/MES/NQ/MNQ RTH split on a 24-hour chart (N1, `rthSource=EquityIndexLegacy`, IP `RthSplit` L643–665) and the fixed Asia/Europe/NY clock windows (N5). - **Every new `.cs`** needs a `PLStudies.csproj` `<Compile Include>` and a one-time PowerLanguage Editor import, listed in the PR (R2; `_InstrumentProfile.Function.CS` is csproj L522 @5c9ccd2). A research draft that remains in `sandbox/` is documented in `docs/research-strategy-header.md` and is not given that Include until the owner approves promotion. - **Error is logged before every throw** that leaves M-INSTR code (`InstrumentError.Fail`, IP L91–98; host `traceAlways`). In M3 a log-only refusal is caught by `InstrumentStreams.Guard` and written once as a Warning instead (§6.2); `MonthMask.Of` is the one accepted exception (§12.4 note). - **No silent fallbacks** (R6), including no weekday business-day fallback.

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1. Relation to `instrument-profile-spec.md` rev 3

1.1 Kept from rev 3

The engine × class support matrix (rev 3 §8.1) and its DECIDED answers: rev-3 Q1 ETF = Stock; rev-3 Q2 FLOW throws on stocks; rev-3 Q3 VPFUT throws on non-futures; rev-3 Q4 no force flag; rev-3 Q5 DVP/G0 Unsupported on FX, VP Degraded (visual) on FX; rev-3 Q7 `profileHash` outside configHash; rev-3 Q8 `_RegimeEngine` on ES/MES only; rev-3 Q9 ES $17; rev-3 Q10 USD only; rev-3 Q11 EVENT Degraded on USD FX pairs. Also kept: the INSTR/CFG log idea, the `EngineNotSupportedException` format (rev 3 §6), the cost-floor formula (IP-8.3).

1.2 Conflicts (this spec wins)

#	Rev 3	Conflict	Rev 2
C1	IP-1.1/IP-2.2 classify by `Category.ToString()`; probe uses `Enum.GetNames`	R1	Exhaustive `switch` on `ESymbolCategory` in the host (§4.2).
C2	IP-1.4 "Build Volume On is not on `IInstrumentSettings`"	Wrong: `Info.BV0` is typed (`_VolumeProfileBase` L1505)	`volumeBasis` from `Info.BV0`.
C3	IP-4.4 "Future = today identity"	Futures path is ES-only	Per-root exchange facts (§8) + chart sessions; ES bit-identical.
C4	IP-8.1 "q is never reduced"	Hard caps reduce today (L2943–2949)	§9 describes the actual code.
C5	IP-8.3 MES $2.47	R9	MES $2.50, floor 20 ticks (integer-cents formula, §9).
C6		R1	

#	Rev 3	Conflict	Rev 2
	PR-I0 `_InstrProbe` uses reflection (`_InstrProbe` L4/67/74/75/144; `_InstrProbeCore` L5/54/126/167)		Typed `_InstrDump` (PR-M0); `_InstrProbe` + `tests/InstrProbeTests` retired in PR-M1e (Q6 DECIDED).
C7	Rev-3 line numbers	Stale	Re-read on `7dbdb1b`.
C8	PR-I4 "no futures holiday re-route"	Other CME product groups have other holidays/early closes	Calendar rule §6.4 S4 and PR-M8; ES keeps `_CmeHolidays`.
C9	Rev-3 session handling = CME equity-index clock for every future	R12	Sessions from `Bars.Sessions` (§4.5); no session times in presets.

2. Purpose

Today an engine run on anything other than @ES/@MES silently uses ES facts, guesses from strings, or fails later with an unclear error. M-INSTR: (1) detects the instrument from typed MC data; (2) takes session times, trading day and session end from the chart's sessions; (3) resolves the exchange facts each engine needs (tick, big point value, currency, contract months, expiry and roll rule, calendar, cluster) from detection plus a small exchange-facts preset per root; (4) applies them, ES bit-identical; (5) logs; (6) refuses clearly under R13.

3. Audit: ES- and futures-specific hardcoding on `origin/main` @ `7dbdb1b`

Method: `rg` over every `*.CS`, `specs/*.md`, `tests/*` for 0.25, 12.5, $50, `"ES"`/`@ES`, H/M/U/Z, `ThirdFriday`, `TimeSpan(9,30)`, (16,0), (16,15), (17,0), (18,0), (15,15), Globex, Eastern/Central, USD, `Category.ToString`, `SymbolRoot`, `PortfolioCluster`, cost literals (471 raw hits, `/workspace/m-instr/grep1.txt`), reduced to one row per distinct hardcoding, plus the 8 rows the rev-1 review found missing (63–70). Column "Sess" = **S** where the row is replaced by chart-session data under R12 (§4.5), **T** where it is a frozen pre-registered trial value (kept, coverage-checked, §6.4 S3).

Engine tags: CS host `_CustomSignal`; VPFUT `_VolumeProfileFuturesBase`; CLOCK `_CmeSessionClock`; CAL holiday calendars; ROLL `_RollDetector` + host roll code; EVENT `_EventCalendar`; G0 `_Gate0`/`_TestGate0`; FLOW `_OrderFlow`; REGGATE `_RegimeGate`/`_Test6`; REGIME `_RegimeEngine`; DAILY `_DailyContext`; TOD `_TodNormaliser` (host side); DVP `_DvpEngine`/`_DvpEngineTest`; SIZE CS-4 sizing; COST; PORT M-PORT; MTF `_MtfMath`; LEVEL, AVWAP, RANGE, WEEKLY, PSTATS `_ProfileStats`; LEGACY pre-engine studies.

3.1 Table

#	file:line	What is hardcoded	Engine	Profile field	Sess
1	`_CustomSignal` L1603–1618	`PortfolioCluster` switch on root strings: EQIDX (L1611), RATES (L1612), ENERGY (L1613), METALS (L1614); others fall through to the raw root	PORT	`cluster`	
2	`_CustomSignal` L2462	`autoTradingDay(barTime, Bars.Info.Category.ToString())`: MC enum → string (R1)	CS	`tradingDay` (typed)	S
3	`_CustomSignal` L2581–2587	`autoTradingDay` substrings "FOREX"/"CURRENCY"/"CRYPTO": FX day = ET+7h; crypto = UTC date	CS	`tradingDay`	S
4	`_CustomSignal` L2587–2615 @d3c0c1a	`exchangeZoneIdOf` token map (PR-M4 replaced the substring `Contains` match): split `Info.Exchange` on non-alphanumerics, exact case-insensitive token match, **first matching token wins**. Chicago → Central: CME, XCME, GLOBEX, CBOT, ECBOT, CBT, XCBT, CBOE. New York → Eastern: NYMEX, COMEX, NYSE, NASDAQ, ARCA. Anything else → Build refuses `tz:exchange`. Composite / feed names (NYSEARCA, NASDAQGS, CMEGLOBEX, AMEX, BATS, CFE) are unmapped pending a live check of real `Bars.Info.Exchange` strings on dxFeed (**A3 OPEN**, §12.4). (History: dump @CL/@GC showed 18:00–17:00 on Exchange and Local, §4.4.2 A3; the NYMEX/COMEX → Eastern entry is unchanged.)	CS	`barZone` / exchange zone for D2 profileHash (§6.1)	S (partly)
5	`_CustomSignal` L29, L2287–2301	`TICK_FALLBACK = 0.01` with a bare `catch {}`	CS	`tick` (throw)	
6	`_CustomSignal` L2062	`_NyseHolidays` created for every instrument	CAL	`calendar`	
7	`_CustomSignal` L3228–3240	`tradingDayCount` uses NYSE for non-stocks (warns once)	CAL	`calendar`	
8	`_CustomSignal` L921	`isRthBar` returns true on the generic base	CS	`rthSplit` (§4.5, N1)	S
9	`_CustomSignal` L949	`todMinuteOfSession` default; futures override = 18:00 ET (row 18)	TOD	`dayOpen`	S
10	`_CustomSignal` L1063–1072	DAILY roll-offset warning assumes the 18:00 ET splice; `Category == Future` only (misses `FutureRolover`, F2)	DAILY	`tradingDay`, `assetClass`	S
11	`_CustomSignal` L1801, L1836	`RollGapTicks = 8` (ES-tuned; `_RollDetector` L12–13)	ROLL	none (Q4)	
12	`_CustomSignal` L1866, L1881–1882	Roll warning / `m_rollGlobexKey` require `Category == Future`; `isFuturesCategory` (`_VolumeProfileBase` L1440–1442) also accepts `FutureRolover`	ROLL	`assetClass`	
13	`_CustomSignal` L1840–1842	Roll inheritance keyed by `_MtfMath.SymbolRoot` (row 31). (Rev 1 cited L1778–1783, which is the HtfStreams same-root throw L1776–1785, also a `SymbolRoot` user, fixed by PR-M1a.)	ROLL	`root`	
14	`_CustomSignal` L749, L766	DVP root = `SymbolRoot(...).ToUpperInvariant()`	DVP	`root`	
15	`_CustomSignal` L2891–2953	`unitNotional` 0 = entry × BigPointValue, implicit USD	SIZE	`bigPointValue`, `currency`	
16	`_CustomSignal` L2994–3002	`roundTripCostPerContract` = 2 × (MC Commission + Slippage)	COST	`cost.roundTrip`	

#	file:line	What is hardcoded	Engine	Profile field	Sess
17	`_CustomSignal` L2008	AVWAP `sessionDate = tradingDayKey`, `rth = isRthBar`	AVWAP	`tradingDay`, `rthSplit`	S
18	`_VolumeProfileFuturesBase` L489–493	TOD minute = minutes since 18:00 ET	TOD	`dayOpen`	S
19	`_VolumeProfileFuturesBase` L52–63	`DEFAULT_*` strings in ET: Asia, Europe, NY, RTH 09:30–16:15, halt 17:00–18:00, Globex 18:00/17:00	VPFUT	session data / N1 / N3	S
20	`_VolumeProfileFuturesBase` L66–77	Same defaults as `TimeSpanS` (VPF-13)	VPFUT	as row 19	S
21	`_VolumeProfileFuturesBase` L361	`tradingDayKey` = `GlobexDate` (18:00 ET)	VPFUT	`tradingDay`	S
22	`_VolumeProfileFuturesBase` L396–398	`DefaultRthValueEnd` 16:00 ET (value window, D5)	VPFUT	`rthSplit` (N1)	S
23	`_VolumeProfileFuturesBase` L472–476	`EOwnProfileWindow { MultiChartsProfile = 0, GlobexDay, Rth }`	VPFUT	names stay; GlobexDay = chart trading day	S
24	`_VolumeProfileFuturesBase` L517; `_CmeSessionClock` L373–379	`isRthBar` = RTH **value** window (09:30, 16:00] ET via `IsRthValueBarEt`; `m_rthEnd` 16:15 is the session flag, not the value window	VPFUT	`rthSplit` (N1)	S
25	`_VolumeProfileFuturesBase` L975–995	Tick-chunk plan per Globex day	FLOW	`tradingDay`	S
26	`_VolumeProfileFuturesBase` L1627, L1635–1671	VPFUT's own `UseCmeHolidayTable` (L1627) and hooks: `IsGlobexClosedDay` L1635, `IsNoRthDay` via NYSE L1641–1646, `HaltEt` L1656, `RthCloseEt` L1665, `_CmeHolidays.Rules.Get(date).Closed` L1671	CAL	`calendar`	
27	`_CmeSessionClock` L48–61	"CME equity-index hours" defaults in ET: RTH, RthValueEnd, halt, Globex open/close, Asia/EU/NY	CLOCK	session data / N1 / N3	S
28	`_CmeSessionClock` L81	Clock's `UseCmeHolidayTable = true` (separate from VPFUT's L1627, row 26)	CAL	`calendar`	
29	`_CmeSessionClock` L248–249	`GlobexTradingDayEt(et)` 18:00 ET open	CLOCK	`tradingDay`	S
30	`_CmeHolidays` L13–45, L65	CME equity-index holiday/early-close rules (13:15 ET L44, Good Friday 09:15 ET L45, halts L65)	CAL	`calendar` (stays; not session times)	
31	`_MtfMath` L29–39	`SymbolRoot`: letters only; drops a trailing month letter. "@6E", "6EZ26" → ""; "EUR/USD" → "EUR" (F1)	MTF	`root`	
32	`_RollDetector` L23	`gapTicks = 8`	ROLL	none (Q4)	
33	`_RollDetector` L43	`RollRuleMondayFrom = 2022-04-01`	ROLL	`roll`	
34	`_RollDetector` L46–51, L56–67	`ThirdFriday`, quarterly `SeriesWindow`	ROLL	`contractMonths`, `expiry`, `roll`	
35	`_EventCalendar` L20	Event rows in ET (correct for US macro)	EVENT	`barZone` for conversion	
36	`_EventCalendar` L128	Default event classes (no EIA for ES)	EVENT	`eventClasses`	
37	`_EventCalendar` L189	`EventRules.RollRuleMondayFrom` 2022-04-01	EVENT	`roll`	
38	`_EventCalendar` L290–303	`ExpiryDate`/`RuleExpiryDate` 3rd Fri; `RollDate` Mon before (2022+) else Thu −8	EVENT	`expiry`, `roll`	
39	`_EventCalendar` L1913–1928	`nextQuarterlyExpiry`, `SessionsToExpiry`, `RuleRollDate`: quarterly	EVENT	`contractMonths`	
40	`_Gate0` L328	`KdeBandwidthTicks = 6`	G0	none (ES canary)	
41	`_Gate0` L358	no `tick` default (M6a). Host sets `tickSize()` (`_TestGate0` L253). 0 / unset: Error line, then throw	G0	`tick`	
42	`_Gate0` L484, L624, L1047	RTH 09:30–16:00 ET, 30-min periods, flat 15:55 (windows not moved by M6a)	G0	frozen trial windows	T
43	`_Gate0` L1185–1186	Doc: "q × widened stop × $50"; `costFloorTicks` only in comments	G0	doc	
44	`_Gate0` L1214	KD-02 passes `root` (host: Data1 `p.root`) into `_DvpEngine`, not a hardcoded `"ES"` (M6a)	G0/DVP	`root`	
45	`_TestGate0` L42, L156–158	"CME Globex" template; $2.25/side + 0.5 tick/side = $17	G0	`cost`	
46	`_OrderFlow` L124	`TickChunkBoundaryEt = 17:00`	FLOW	`sessionEnd`	S
47	`_OrderFlow` L935	`PlanGlobexDays(...)`	FLOW	`tradingDay`	S
48	`_RegimeGate` L45–48	A2-63: rthOpen 09:30, activeEnd 15:30, exit 15:55, rthClose 16:00 ET	REGGATE	frozen trial windows	T
49	`_RegimeGate` L55–62	Roll years, digest `fdd63bd8f179a8d4` (L60), `tradingDay = "Globex18ET"` (string, L62)	REGGATE	`roll`, typed key	
50	`_Test6` L63, L89, L105–106	ET zone string, Globex key, `RollDetection.ToString()` (repo enum) into the hash	REGGATE	typed key	S
51	`_DailyContext` L30, L33, L38	defaults `{252,126,63}`, `hvAnnualisation 252`; `ApplySessionsPerYear` replaces those defaults with `spy`, `spy/2`, `spy/4`	DAILY	`sessionsPerYear`	

#	file:line	What is hardcoded	Engine	Profile field	Sess
52	`_RegimeEngine` L63, L274	`sessionsPerYear` default 252; `sqrt(63/sessionsPerYear)` (the 63-session window stays)	REGIME	`sessionsPerYear`	
53	`_LevelStudy` L80	no `tickSize` default (M6d). Host sets `tickSize()` (CS L1477). Gate 0 sets `study.tickSize = tick` (L471)	LEVEL	`tick`	
54	`_AvwapManager` L154	no `tickSize` default (M6d). Host sets `tickSize()` (CS L2147). Configure rejects a tick that is not > 0 (L196)	AVWAP	`tick`	
55	`_RangeProfile` L31	`ringBars = 2880` (a **bar** count; M6e literal kept)	RANGE	none: literal kept	
56	`_WeeklyProfiles` L19, L26	Default source `RthSessions`. S7 when the filled profile has no `rthSplit` (M6e). `isRthBar` filter unchanged	WEEKLY	`rthSplit` (N1)	S
57	`_PortMM` L388, L740; `_PortfolioRiskManager` L87, L129	USD-only book (typed `eMTPA_MCSC_USD`, correct)	PORT	`currency`	
58	`_DvpEngineTest` L213, L230	`root == "ES"`: roll table ES-only	DVP	`roll`	
59	`_DeltaDivRevFuturesLELX` L152, L175, L528–564	Globex state, Globex-day trade limit	LEGACY	note only	
60	`_ExitHalfRiskHalfRoll` L107–113	"One tick" = `1 / PriceScale * BigPointValue` (0.5 on ES if PriceScale = 100: neither a tick nor $/tick) (F3a)	LEGACY	`tick`	
61	`_EconomicCalendar` L231; `_EhlersFilter` L149; `_StructuralTargetPatterns.Strategy` L1060; `_GenerateNNTrainingData` L354	FX pip = `PointValue × 10 × BigPointValue`	LEGACY	note only	
62	`_EpisodicPivotBase` L32–35; `_ParabolicShortSESX` L51–52; `_BreakoutAutoLELX` L42; `_TrendLine` L2755–2756; `_VolumeProfileBaseLog` L206, L426	US-stock 09:30/16:00 ET; `m_tick = 0.25`	LEGACY	note only	
63	`_CustomSignal` L1888–1889	`findTz("Eastern Standard Time")` + `GlobexTradingDayEt` for the roll key on any future	CS	`tradingDay` (removed)	S
64	`_DvpEngineTest` L119–121; `_Test6` L100–102	Host-generated quarterly `RollDates` (`for m = 3..12 step 3: RollDates.Add(EventRules.RollDate)`)	DVP	`contractMonths`, `roll`	
65	`_ProfileStats` L47	`ringBars = 1440` (bar count; M6e literal kept)	PSTATS	none: literal kept	
66	`_ProfileStats` L575	`disagree` when	mcPoc − cvPoc	> 4 × tick (ES-tuned). CFG note when the engine is enabled (M6e)	PSTATS
67	`_TestGate0` L157	NQ slippage $2.50 in the canary note	G0	`cost` (doc)	
68	`_MtfMath` L28	`SymbolRoot` doc says returns "@ES"; code returns "ES"	MTF	doc fix	
69	`_DvpEngine` L2445–2447	`RootError`: empty == empty is a match, so any two digit-led roots pass (F1)	DVP	`root`	
70	`_ExitHalfRiskHalfRoll` L123	`GenerateStopLoss(m_firstStopPrice)`: a price distance passed where MC takes currency (F3b)	LEGACY	`bigPointValue`	

Counts. 70 rows; each counted in exactly one area. Rows 61 and 62 group several legacy files each.

Area	Rows	Count
CS host	2, 3, 4, 5, 8, 9, 10, 63	8
VPFUT	18–24	7
CLOCK and CAL	6, 7, 26–30	7
ROLL	11, 12, 13, 32–34	6
EVENT	35–39	5
G0	40–45, 67	7
FLOW	25, 46, 47	3
REGGATE	48–50	3
DAILY and REGIME	51, 52	2
SIZE and COST	15, 16	2
PORT	1, 57	2

Area	Rows	Count
DVP	14, 58, 64, 69	4
MTF	31, 68	2
AVWAP	17, 54	2
LEVEL, RANGE, WEEKLY, PSTATS	53, 55, 56, 65, 66	5
LEGACY	59–62, 70	5
Total		**70**

Tests. ES literals by project: VpHelper 102, OrderFlow 60, Gate0 45, Roll 39, PortMM 23, Dvp 16, Gmm 15, PriceAction 10, fewer elsewhere. These are ES fixtures by design and stay as the bit-identity guard; M-INSTR adds per-root fixtures. Exception: Gate0Tests is edited in PR-M6a (tick, §7).

3.2 Bugs today (fixed independently, PR-M1a–c)

- **F1 (wider than rev 1).** `SymbolRoot` drops digit-led roots ("@6E", "6EZ26" → "") and truncates FX pairs ("EUR/USD" → "EUR", colliding across EUR pairs). `_DvpEngine.RootError` L2445–2447 then treats "" == "" as a match, so any two digit-led roots pass the same-root check, and roll inheritance (L1840–1842) and the HtfStreams check (L1776–1785) are wrong for them. Fix in PR-M1a (§4.3); `RootError` treats an empty root as an error.
- **F2.** `Category == ESymbolCategory.Future` at L1069, L1866, L1882 misses `FutureRolover`; use the host's `isFuturesCategory` (`_VolumeProfileBase` L1440–1442). PR-M1b.
- **F3 (two bugs).** (a) `_ExitHalfRiskHalfRoll` L107–113 computes `1/PriceScale*BigPointValue`, which is neither a tick nor $/tick; (b) L123 passes `m_firstStopPrice`, a price distance, to `GenerateStopLoss`, which takes a currency amount. Fix: tick = `MinMove/PriceScale`; amount = distance × `BigPointValue` (per contract, matching the strategy's existing per-contract/position mode, VERIFY in the PR). PR-M1c.
- **F4.** `TICK_FALLBACK = 0.01` (L29) with a bare `catch {}` in `resolveTick` (L2287–2301) → Error + throw. PR-M1b.

3.3 MC-enum string uses on main (findings for PR-M1b / allow-list for MI-T14)

file:line	Use	R1 status	Action
`_CustomSignal` L2462	`Category.ToString()` passed to `autoTradingDay` substring tests	violation	PR-M1b: typed `switch` (`ClassOf`)
`_InstrProbe` L67, L74, L75	`Category.ToString()` / reflection	violation	PR-M1e: retire the probe
`_CustomSignal` L279	`TimeZone.ToString()` into a log line	allowed (log)	none
`_CustomSignal` L1883	`res.ToString()` (`EResolution`) into a Warning text	allowed (log)	none
`_Test6` L105	`RollDetection.ToString()` (repo enum `ERollDetection`, `_CustomSignal` L1791) into the A2-63 hash	outside R1 (repo enum) but hash-fragile	PR-M6h: explicit switch, same spellings

4. Detection (typed MC API)

4.1 Members used

Status: **confirmed-PG** = in the MultiCharts .NET Programming Guide (v1.1 §4.2.3 "Properties of the Data Series", pp. 21–22; `mc-sessions-research.md` §6); **compiles** = used in code on main (so the member exists); **VERIFY** = neither, or a behaviour (not the member) is unconfirmed.

Member	Type	Status	Source
`Info.Category`	`ESymbolCategory`	compiles; values Future 0, FutureOption 1, Stock 2, StockOption 3, Index 4, CurrOption 5, MutualFund 6, MoneyMarketFund 7, IndexOption 8, Cash 9, Bond 10, Spread 11, Forex 12, CPCSymbol 13, Composites 14, FutureRoot 15, FutureOptionRoot 16, StockOptionRoot 17, FutureRolover 18, Commodity 19, Crypto 20, Warrant 21, None 254 from Ettienne's dump3 run (box copy, not in repo)	`_VolumeProfileBase` L1442, `_CustomSignal` L1069
`Info.MinMove`, `Info.PriceScale`	double	compiles, PG	`_CustomSignal` L2291–2292
`Info.BigPointValue`	double	compiles, PG ("in the instrument currency")	`_CustomSignal` L1660, L1678, L2360
`Info.PointValue`	double	compiles, PG (a **different** member from BigPointValue)	`_EconomicCalendar` L231
`Info.Description`	string	compiles, PG	

Member	Type	Status	Source
			`_EconomicCalendar` L243, L679
`Info.Exchange`	string (data value; parsing allowed)	compiles, PG	`_CustomSignal` L2601–2603
`Info.Name`	string (data value)	compiles, PG	`_CustomSignal` L1886
`Info.Resolution.Type` / `.Size`	`EResolution` / int	compiles, PG	`_CustomSignal` L1069, L966, L2245
`Info.TimeZone`	`RequestTimeZone` {Local, Exchange, GMT}	compiles, PG	`_CustomSignal` L279, L2594–2606
`Info.BVO`	`BuildVolumeOn`	compiles	`_VolumeProfileBase` L1505
`Info.SessionName`	string	**confirmed-PG**; log only, never parsed. Returns the template name as set on the chart: `Default` for Default, `CME: Equity Index Futures [E-mini] (GLOBEX)` for the GLOBEX template (**CONFIRMED** `01M6HSDCPGEWFS9VAYZG2C808V.txt`, `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`)	PG §4.2.3, dump
`Info.Request`	`DataRequest`	confirmed-PG; not used by M-INSTR	PG §4.2.3
`Bars.LastBarInSession`	bool	confirmed-PG (§4.2.2). **Not a reliable session-end signal on its own**: not true at the GLOBEX 15:15 break (`01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`); see R14	PG, dump
`Bars.Sessions`	`IROList<SessionObject>` (`.Count`, `[i]`)	**compiles** (`_CheckData` L54 in StartCalc, L198–200; `_QullamaggieBase` L189–192; #88 head `mtfSessions` via `IInstrument`), so the members exist; **not in the PG** (forum/staff only). Populated at StartCalc and identical on the first CalcBar: **CONFIRMED** in all 13 runs (`sameAsStartCalc(yes)`, §4.4.2)	main, #88 head, dump
`SessionObject.StartDay/EndDay` (`DayOfWeek`), `.StartTime/.EndTime` (`TimeSpan`)		compiles (same sites); zone of the times = the chart (Format Instrument) time zone of the stream: **CONFIRMED** (@ES 17:00–16:00 on Exchange `01M6HSDCPGEWFS9VAYZG2C808V.txt`, 18:00–17:00 on Local `01M6HSDBFJ9WRF71G51B82VKCH.txt`; bar times shift with it)	main, dump
`SessionObject.EndOfSession(DateTime)`, `Sessions.SessionForDate(DateTime)`, an end-of-session bool		forum only: **VERIFY**; M-INSTR does **not** use them	research §6
`Info.ASymbolInfo2`	`MTPA_MCSymbolInfo2` (COM struct)	**confirmed-PG**: listed under §4.2.3 "Instrument properties" (`MTPA_MCSymbolInfo2 ASymbolInfo2 { get; }`, pdftotext `/workspace/research/pg11.txt` L901); struct fields in the `MTPA_MCSymbolInfo2` section (L3828–3867, PG p. 82): `SymbolName`, **`SymbolRoot`** ("symbol root (for futures), for example ES, CL", L3833), `SymbolDescription`, `SymbolExchange`, `SymbolCategory` (`MTPA_MCSymbolCategories`), **`SymbolCurrency`** (`MTPA_MCSymbolCurrency`), **`SymbolExpiry`** (`DateTime`, "expiration date (for futures)"), `MinMove` (int), `PriceScale` (double; PG example "for ES = 1/100", L3853, whereas `Info.PriceScale` dumped `100`), `BigPointValue`, `PutOrCall`, `StrikePrice`, **`UnsupportedFields`** (`MTPA_SymbolFields`, "fields which are not provided by the source"); "received through ... `Bars.Info.ASymbolInfo2` for the current symbol" (L3867). **CONFIRMED** (Ettienne `_RootDump` 2026-11-03, relayed by Coder, dxFeed; @ES, @CL, @6E, SPY, table in §4.4.3): `SymbolRoot`, `SymbolCurrency`, `SymbolCategory`, `SymbolExchange`, `MinMove`, `PriceScale` (= 1/`Info.PriceScale`: 0.01 vs 100; 6E 1E-05 vs 99999.999999999985) and `BigPointValue` (= `Info.BigPointValue`); `SymbolName` = `Info.Name` in all four; `UnsupportedFields` = 0 in all four. `SymbolExpiry` is a **placeholder** on continuous futures (2038-01-01) and stocks (1900-01-01) (§4.4 B1). `MTPA_SymbolFields` is **not** `[Flags]` but its values are powers of two: `eMTPA_SF_Root`=1, `Exchange`=4, `Description`=8, `Category`=16, `Expiry`=32, `PutOrCall`=64, `StrikePrice`=128, `Multiplier`=256, `Currency`=512, `ALL`=-1. Credit: Ettienne, rev 2.3. **Rev 2.4: `MTPA_SymbolFields.eMTPA_SF_Root` and `MTPA_MCSymbolCurrency.eMTPA_MCSC_USD` CONFIRMED (compile)**: Ettienne's MI-T1 PLEditor Compile All of `5c9ccd2` passed on 2026-11-04 (the first compile failed only because `_InstrumentProfile.Function.CS` had not yet been imported; one-time import), so both members exist as the code uses them typed (IP `InstrumentMcFields.RootUnsupported` L973, `CurrencyCode` L977). The code never relies on the numeric value (the value 1 is still only the test mirror, `tests/InstrumentTests/gen.sh` L23). The MC runs printed `rootSrc=mc` on 6E, ES and AAPL (root not flagged unsupported)	PG §4.2.3, PG p. 82
`Info.Expiration`	`DateTime`	confirmed-PG as a property; never compiled in the repo (the probe reads it by reflection): **VERIFY**, and **not needed**: expiry comes from `SymbolExpiry` on dated contracts, else the roll calendar/detector (§4.4 B1); not used by M-INSTR	PG §4.2.3
			main

Member	Type	Status	Source
`StrategyCurrencyCode`	`MTPA_MCSymbolCurrency`	compiles on an `IStrategy` from a portfolio signal only (`_PortfolioRiskManager` L125–129); `_TestPortMM.Signal.CS` (which also used it at L90) was **deleted in PR-M4** (A2). On a plain `SignalObject`: **VERIFY**, and **not needed**: currency comes from `ASymbolInfo2.SymbolCurrency` (CONFIRMED, §4.4 B2); not used by M-INSTR	
Per-symbol currency on `Info`	—	no documented member; none guessed. Currency comes from the preset (USD, Q8)	—
Exchange time-zone member, `SessionStartBar`/`SessionEndBar`, "is Default" flag	—	not documented; not used	research §6
Holidays from code	—	**not available** (MC staff, 2013/2015/2022)	research §6

4.2 Classification (host, exhaustive switch; replaces rev-1 MI-T1)

```
static EAssetClass ClassOf(ESymbolCategory c) {
  switch (c) {
    case ESymbolCategory.Future: case ESymbolCategory.FutureRolover: return EAssetClass.Future;
    case ESymbolCategory.Stock: return EAssetClass.Stock;      // ETF = Stock (rev-3 Q1)
    case ESymbolCategory.Forex: return EAssetClass.SpotFx;
    case ESymbolCategory.Crypto: return EAssetClass.Crypto;
    case ESymbolCategory.Index: return EAssetClass.Index;
    case ESymbolCategory.FutureOption: case ESymbolCategory.StockOption: case ESymbolCategory.CurrOption:
    case ESymbolCategory.IndexOption: case ESymbolCategory.MutualFund: case ESymbolCategory.MoneyMarketFund:
    case ESymbolCategory.Cash: case ESymbolCategory.Bond: case ESymbolCategory.Spread: case ESymbolCategory.CPCSymbol:
    case ESymbolCategory.Composites: case ESymbolCategory.FutureRoot: case ESymbolCategory.FutureOptionRoot:
    case ESymbolCategory.StockOptionRoot: case ESymbolCategory.Commodity: case ESymbolCategory.Warrant:
    case ESymbolCategory.None: return EAssetClass.Unsupported;
    default: throw new InvalidOperationException("M-INSTR: unmapped ESymbolCategory " + ((int)c).ToString(CultureInfo.InvariantCulture)
      + " (not in spec §4.1); update ClassOf (§4.2) for this MC build.");
  }
}
```

All 23 members are listed by name; the `default` throws on a value added in a later MC build (formatting the int is a log, not a parse). It compiles only in PowerLanguage Editor (no test project references PowerLanguage), so it is checked by the PLEditor compile in PR-M2/M3, not by a test. The MC-free core uses only `EAssetClass`.

As merged (rev 2.4, @5c9ccd2). `ClassOf` stays in `_CustomSignal` (CS L2492–2510, region `<instr-host-core>`); `EAssetClass` lives in IP (`<instr-asset-class>`). Two callers:

- `buildProfiles()` calls `InstrumentMcFields.Read(info, ClassOf(info.Category))`. An unmapped `ESymbolCategory` (the `default` throw of `ClassOf`) is **Error + throw `class`** under M4 (A1), same as every other Build refusal. (M3: it was a log-only Warning and a null profile; that path is gone.)
- `classOfData1()` (CS L2708–2723, `TradingDayRoll = Auto`) already writes the Error line (`traceAlways`) and rethrows.

4.3 Root resolution (PR-M1a, fixes F1; rule rewritten in rev 2.2)

1. **Clean:** trim; remove a leading `@`; remove a `#…` continuous suffix; uppercase invariant; remove every `/` ("EUR/USD" → "EURUSD"). In PR-M1a (`_MtfMath.SymbolRoot`, MC-free, no class) the `/` removal applies to every name; **PR-M2 gates it to Spot FX** (`SpotFx`) once the resolver knows the class.
2. **Root:** take the leading digits plus the run of letters that follows them (stop at the first character that is neither). If the letter run has **≥ 2 letters**, its last letter is a month code (`FGHJKMNQUVXZ`) and the character right after the run is a digit, drop that last letter. Nothing else is removed or kept: exchange suffixes (`:XCME`), spaces, year digits and .-classes fall away because the run stops there. Examples (MI-T2): `ESZ2024` → `ES`; `'ES 12-24'` → `ES`; `ESZ24:XCME` → `ES`; `ESZ4` → `ES`; `CLZ2026` → `CL`; `MESZ26` → `MES`; `MESH27` → `MES`; `@6E` → `6E`; `6EZ26` → `6E`; `6B` → `6B`; `EUR/USD` → `EURUSD`; `BRK.B` → `BRK`; `Z4` → `Z` (one-letter run: no month drop). (Rev 2.1 steps 3–4, "name minus a trailing month + 1–2 digit year", left `ESZ2024` unchanged; the code on `6853398` (main before #90) L109–119 already used the letter-run form but drops leading digits and does not remove `/`.)
3. **MC's stored root is primary (PR-M2 resolver; rev 2.3, Ettienne; CONFIRMED by Ettienne `_RootDump` 2026-11-03, relayed by Coder).** MC stores the root itself: `Bars.Info.ASymbolInfo2.SymbolRoot` (PG §4.2.3, `/workspace/research/pg11.txt` L901; struct L3828–3867, field L3833). dxFeed **populates it for futures and stocks** (@ES → `ES`, @CL → `CL`, @6E → `6E`, SPY → `SPY`; §4.4.3). The M2 resolver takes it as the **primary** root (trimmed, uppercase invariant); the #90 parser (`_MtfMath.SymbolRoot`, steps 1–2) is a **pure fallback**, used only when `(s.UnsupportedFields & MTPA_SymbolFields.eMTPA_SF_Root) != 0` (typed; also covers `ALL` = -1; `MTPA_SymbolFields` is not `[Flags]` but its values are powers of two, so the bitwise test is valid) **or** `string.IsNullOrWhiteSpace(s.SymbolRoot)`. Never a string or `ToString()` test on the enum (R1). When both roots are non-empty and `!_MtfMath.SameRoot(mcRoot, parsedRoot)`, log one **Warning** `INSTR root-crosscheck mc(<mcRoot>) parsed(<parsedRoot>) name(<Info.Name>)`; **MC's root wins** (it feeds presets, roll inheritance and the same-root checks). The INSTR line records `rootSrc=mc|parsed`. PR-M1a (`_MtfMath`, MC-free, #90) keeps the parser only; it has no access to `Info`. **As merged (rev 2.4):** `InstrumentRoot.Pick` (IP L308–322) implements the choice (MC root trimmed + uppercase; for Spot FX the `/` rule is applied to MC's root too); `Resolve` (IP L298–304) refuses `unknown-field:root` when both are empty. The Warning is logged once at StartCalc (`InstrumentStreams.Start`, IP L934). **Closed 2026-11-04:** the fallback test `(UnsupportedFields & eMTPA_SF_Root) != 0` compiles in PLEditor (MI-T1, §4.1) and the MC runs read `rootSrc=mc` on 6E, ES and AAPL. The parsed-fallback branch itself was not exercised in MC (no symbol flagged the root unsupported); it is covered by the MC-free tests. **MC-internal cross-check (every stream, before presets):** `s.PriceScale × Info.PriceScale = 1` within ±1e-9 relative (ES/CL/SPY 0.01 × 100; 6E 1E-05 × 99999.999999999985 = 0.99999999999999985) and `s.BigPointValue == Info.BigPointValue`. On a mismatch: Error + throw `mismatch:mcPriceScale` / `mismatch:mcBigPointValue` with a clear message, e.g. `M-INSTR: <ENGINE> cannot`

`run on @6E (class Future, root 6E, sessions Default): mismatch:mcPriceScale (ASymbolInfo2.PriceScale 1E-05 × Info.PriceScale 99999.999999999985 = <x>, expected 1 ±1e-9). Check the symbol's settings in QuoteManager.` (§6.3). `MinMove` is equal in all four rows and is logged, not thrown on.

4. **Class-aware handling (PR-M2 resolver only):** stocks keep their share-class suffix (`BRK.B` is its own symbol; the M1a root `BRK` is used only for the MTF/DVP same-root checks), and the `/` gate of step 1 applies to Spot FX only. Presets (PR-M2) match the step-2 root exactly; the rev 2.1 longest-prefix search is dropped because step 2 already isolates the root (`MES` ≠ `ES`, `6E` ≠ `6B`).
5. Other classes when the fallback applies (MC gives no root or flags it): the cleaned name with the same step 2 (crypto `BTCUSD` → `BTCUSD`, `AAPL` → `AAPL`).
6. An empty root (MC and parser both empty) is an error (`RootError` and roll inheritance never compare empty roots).
7. Presets match on root **and** class ("ES" as a stock never gets the ES futures preset).
8. **Cross-check (futures with a preset):** detected tick = `MinMove/PriceScale` and `BigPointValue` equal the preset within 1e-9 relative, else Error + throw `mismatch:tick` / `mismatch:bigPointValue`. The tolerance covers MC's binary priceScale: @6E reports `priceScale(99999.999999999985)` with `minMove(5)` (`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`), raw tick 5 / 99999.999999999985 = `5.0000000000000009E-05` as .NET Framework `ToString("R")` prints it (17 digits; .NET 8 prints `5.000000000000001E-05`) vs the preset 0.00005, relative error ≈ 1.5e-16 (**CONFIRMED** covered; rev 2.3's `…07e-5` was wrong). **After the cross-check passes the profile carries the preset's exact tick and bigPointValue** (IP L792–793, preset values chosen just above), and `tickValue = tick × bpv` (0.00005 × 125000 = 6.25); without a preset it carries the detected tick and `Info.BigPointValue`. Profile facts on the INSTR line and in the profileHash canon are formatted with `InstrumentText.Fact` = `G15`, invariant (IP L111), so @6E prints `tick(5E-05)` on both runtimes; `InstrumentText.Num` = `"R"` (IP L108) is used **only in diagnostics** (mismatch/refusal texts), whose digits may differ between .NET Framework and .NET 8. The cross-check order and which checks are enforced in M3: §6.2. A detected tick ≤ 0 (or PriceScale ≤ 0) with a preset is `mismatch:tick` (enforced), not `tick`. @ES 25/100 bpv 50, @MES bpv 5, @CL 1/100 bpv 1000, @GC 1/10 bpv 100, @6E bpv 125000 all equal the §8.1 presets.
9. **No preset:** roll rule, contract months, expiry and cost are Unknown; an engine that needs one throws (§6.4 S5).

4.4 VERIFY dump `_InstrDump.Strategy.CS` (PR-M0; typed, no reflection)

Ettienne runs it once on @ES (his live chart, Default sessions), @ES on the "CME Globex" template (the D9a/canary chart), @MES, @CL, @GC, @ZN, @6E, one FX pair, one crypto symbol and one stock, and once in Portfolio Trader. Output: see **§4.4.1** (same directory as the standard MCSignal log, N6 DECIDED). Doubles via `ToString("R", CultureInfo.InvariantCulture)`. Restart MultiCharts after changing any symbol's sessions in QuoteManager (cached until restart, §4.5 setup note).

4.4.1 Output file (N6 DECIDED)

- **Directory = the directory of the standard log file the host already creates.** The resolver on `7dbdb1b` is in `_CustomSignal.Strategy.CS`, nested `LogFileSingleton` constructor **L71–91**: `Path.Combine(Environment.GetFolderPath(SpecialFolder.LocalApplicationData), "MCSignalLogs")` (L79), `Directory.CreateDirectory` (L80), a zero-length write to prove access (L82), and on any exception the fallback `Path.GetTempPath()` (L85–88). The resolved file is exposed as `LogFileSingleton.FilePath` (L107) and obtained via `LogFileSingleton.GetInstance` (L93; used at L371 and L395). The same two-step rule is duplicated inline in `TradeJournal` (constructor L168–184, dir L176, fallback L182, `MCJournal_`/`MCEvents_` files).
- **Reuse, no new path.** `_InstrDump` derives from `_CustomSignal` (precedent: `_MtfProbe : _CustomSignal`, `_MtfProbe.Strategy.CS` L89). PR-M0 adds one accessor to `_CustomSignal`, no behaviour change: `protected string LogDirectory() { if (m_logger == null) m_logger = LogFileSingleton.GetInstance(null); return Path.GetDirectoryName(m_logger.FilePath); }` (same lazy pattern as `traceAlways`, L395). It therefore returns the directory of the log file actually created, including the `%TEMP%` fallback. No `const` path, no `C:\Share` path, no new environment lookup in `_InstrDump`.
- **Name**, following the existing `MCSignal<yyyyMMdd-HHmmss>-<pid>.csv` (L76) / `MCJournal_<runId>.csv` / `MCEvents_<runId>.csv` pattern: `MCInstrDump_<symbol>_<yyyyMMdd-HHmmss>-<pid>.txt`, timestamp `DateTime.Now` with `CultureInfo.InvariantCulture`, `<symbol>` = `Bars.Info.Name` with every `Path.GetInvalidFileNameChars()` character and `/` replaced by `_` (`@ES` stays `@ES`; `EUR/USD` → `EUR_USD`). One file per symbol per run (Portfolio Trader: one per symbol).
- **Writing:** one `File.AppendAllText(path, line + Environment.NewLine, new UTF8Encoding(false))` per line (UTF-8 without BOM, as the log, L66), doubles `ToString("R", CultureInfo.InvariantCulture)`. On the first line `_InstrDump` writes one trace line to the standard log, `INSTRDUMP file(<full path>)` (same form as `JOURNAL file(...)`, L2407), and one `Output.WriteLine` with the path. IO errors go to Output and never stop the strategy.

- **Collecting (PowerShell one-liners, no variables).** List the dumps in both possible directories, newest last: `Get-ChildItem -LiteralPath (Join-Path ([Environment]::GetFolderPath('LocalApplicationData')) 'MCSignalLogs'), ([System.IO.Path]::GetTempPath()) -Filter 'MCInstrDump_*.txt' -ErrorAction SilentlyContinue | Sort-Object -Property LastWriteTime | Format-Table -Property LastWriteTime, Length, FullName -AutoSize` Make a Desktop folder for them: `New-Item -ItemType Directory -Force -Path (Join-Path ([Environment]::GetFolderPath('Desktop')) 'InstrDump')` Copy every dump into it: `Get-ChildItem -LiteralPath (Join-Path ([Environment]::GetFolderPath('LocalApplicationData')) 'MCSignalLogs'), ([System.IO.Path]::GetTempPath()) -Filter 'MCInstrDump_*.txt' -ErrorAction SilentlyContinue | Copy-Item -Destination (Join-Path ([Environment]::GetFolderPath('Desktop')) 'InstrDump') -Force` Then send the `InstrDump` folder (or attach the files to the PR-M0 thread); they are committed as `tests/InstrumentTests/fixtures/MCInstrDump_*.txt` (data only; nothing in M1 depends on them).

- **A1 (compiles today):** `name, cat (log), catInt, exch, tz (log), minMove, priceScale, bpv, bvo (log), res.Type (log), sessionName`. Description, PointValue and Resolution.Size are already used in compiled code (`_EconomicCalendar` L231/L243/L679, `_CustomSignal` L966/L2245) and are not dumped (`Info.SessionName`, confirmed-PG; record whether it prints "Default" or the template name).

- **A2 Sessions at StartCalc vs first bar:** print `Bars.Sessions` (`n`, then each `StartDay StartTime -> EndDay EndTime`) **in StartCalc and again on the first CalcBar**, and the first bar's `Bars.Time[0]` and `Bars.LastBarInSession`. Records: populated at StartCalc (yes/no/null), identical at the first bar, multiple sessions per day kept or collapsed (forum: fixed in MC .NET 9.0). **Result: CONFIRMED** populated and identical in all 13 runs; multiple sessions per day kept (GLOBEX: 9 entries, 17:00–15:15 and 15:30–16:30 each weekday).
- **A3 Session and bar zone:** run @ES and @CL each twice, Format Instrument → Time Zone = **Exchange**, then **Local**. Compare the same bar's `Time[0]` and the printed session times between runs: (i) whether `SessionObject` times follow the chart zone (staff: "main data series" zone); (ii) the exchange offset for NYMEX/COMEX (settles row 4, L2621). **Result: CONFIRMED** (i): session and bar times follow the chart zone (@ES 17–16 Exchange, 18–17 Local); (ii) @CL and @GC show the same 18:00–17:00 either way, see row 4.
- **A4 Multi-series / Portfolio:** with Data2 on a different template, print Data2's sessions (`BarsOfData(2).Sessions`) to confirm per-series sessions; in Portfolio Trader, print each symbol's sessions and `Info.TimeZone` (PT time zone is one global setting). **Result:** per-series sessions **CONFIRMED** (Data1 Default n=5, Data2 GLOBEX n=9, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`); Portfolio Trader: the PR-M0 dump was incomplete (only @ES written, ds2–10 absent, `01M6HSDBFH56AB94QP1SZ4NFP7.txt`); **CLOSED 2026-11-04** by the PT per-symbol test on `5c9ccd2` (below).
- **B (resolved by `_RootDump`, rev 2.3; Ettienne `_RootDump` 2026-11-03, relayed by Coder):**
- **B1 expiry:** `ASymbolInfo2.SymbolExpiry` is a **placeholder** on continuous futures (2038-01-01 on @ES, @CL, @6E) and stocks (1900-01-01 on SPY). It is used **only when strictly after 1900-01-01 and strictly before 2038-01-01** (i.e. a dated contract); otherwise expiry and roll come from the roll calendar (§8.1) and the roll detector. `Info.Expiration` stays VERIFY and is not needed.
- **B2 currency:** `ASymbolInfo2.SymbolCurrency` (typed `MTPA_MCSymbolCurrency`): **CONFIRMED** USD for all four; non-USD throws (Q8). `StrategyCurrencyCode` is not needed.

- **B3 cross-checks:** `SymbolRoot`, `SymbolExchange`, `SymbolCategory`, `MinMove`, `PriceScale`, `BigPointValue` **CONFIRMED** (§4.4.3); `PriceScale` convention: struct = 1/`Info.PriceScale` (the PG "for ES = 1/100" example, L3853, is the struct's); the resolver check is §4.3 step 3. `SymbolCategory` is typed (`MTPA_MCSymbolCategories`) and is logged, not used for classification (§4.2 stays on `Info.Category`).
- `_RootDump.Strategy.CS` is on main (added in `e1f5b11` when `_InstrDump.Strategy.CS` was retired; csproj entry kept). Its four rows are the MI-T2/MI-T3 fixtures (plain strings/ints/doubles in the MC-free core). The original Ettienne dump of 2026-11-03 (relayed by Coder) is the evidence in §4.4.3; the committed probe reprints the same typed fields.

`{i.Category}` etc. are formatted only for the human log; nothing parses them (R1).

4.4.2 PR-M0 dump results (Ettienne's runs, 2026-11-03 17:28–17:43 ET)

Thirteen fixture files from the PR-M0 `_InstrDump` build (rev 2.3: the 6E rerun `01M6HTKV3SV1Q5QDXBFTCH8D3K.txt` replaces the discarded `01M6HSDCCJZXJHNB165KSF6Z5B.txt`; the latter is still listed in `index.txt` but is not committed), indexed in `/workspace/instrdump/index.txt` (to be committed as `tests/InstrumentTests/fixtures/`; the file names are the upload names, not the `MCInstrDump_*` names). Not run yet: @ZN, an FX pair, a crypto symbol (B1/B2 resolved by `_RootDump`, §4.4.3).

File	Run	Facts read from it
`01M6HSDCPGEWFS9VAYZG2C808V.txt`	@ES, Exchange, Default	n=5, Sun 17:00 → Mon 16:00 … Thu 17:00 → Fri 16:00; same at StartCalc and first bar; session-end bars 16:00
`01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`	@ES, Exchange, GLOBEX template	`sessionName(CME: Equity Index Futures [E-mini] (GLOBEX))`; n=9 (17:00–15:15 + 15:30–16:30); **session-end bars only at 16:00, none at 15:15** (R14)
`01M6HSDC526NQSHVTNJG9AQ7A7.txt`	@MES, Exchange, Default	as @ES, bpv 5
`01M6HSDDB20HAY23SAABKQ0MQ5.txt` / `01M6HSDCMNQG1HPR0B9XQWC4S2.txt`	@CL Exchange / Local	18:00–17:00 in both; bars end 17:00 in both; minMove 1, priceScale 100, bpv 1000
`01M6HSDC20TJ572NEKZBKKRBWD.txt`	@GC, Exchange	18:00–17:00, bars end 17:00; minMove 1, priceScale 10, bpv 100
`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`	@6E, Exchange (rerun 2026-11-03 18:05 ET, after switching @6E to the Exchange & ECN sessions and restarting MC)	`cat(Future)`, exch CME, priceScale 99999.999999999985, minMove 5, bpv 125000; Default session Sun–Thu 17:00 → next day 16:00 (Exchange), first bar 17:06, session-end bars at 16:00 (`lastBarInSession(True)`): **matches ES**. Chart has `bvo(TickCount)`: a futures chart on Tick Count must throw under R7 (volume basis), so this fixture is for **instrument and session facts only**, never a volume-engine fixture. Replaces `01M6HSDCCJZXJHNB165KSF6Z5B.txt` (misconfigured custom sessions, **discarded**; not a fixture)
`01M6HSDBFJ9WRF71G51B82VKCH.txt`	@ES, Local	18:00–17:00; first bar 18:07, session ends 17:00 (= Exchange run + 1 h)
`01M6HSDBG7NET9EVPFJ8Z43MR5.txt`, `01M6HSDBWZ22Q03T3AV5GWP4CY.txt`, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`	@ES Data1 Default + @ES Data2 GLOBEX (three identical runs apart from run id)	Data1 n=5, Data2 n=9 (per-series sessions); Data1 16:00 bar of 2026-10-26 logged as a session end twice (R14 idempotence)
`01M6HSDBFH56AB94QP1SZ4NFP7.txt`	Portfolio Trader, @ES + @CL, Local	`maxDataStream(10)`, ds2–10 `absent`; only the @ES dump was written (superseded: PT per-symbol test passed 2026-11-04)
`01M6HSDC62GCMK5ZANKB61PD5T.txt`	SPY, Exchange	`cat(Stock) catInt(2)`, exch ARCA, `bvo(TickCount)`, Mon–Fri 04:15–20:00, session ends 20:00

CONFIRMED: (1) `Bars.Sessions` is filled at StartCalc and identical on the first bar (all 13); (2) session and bar times follow the chart time zone (@ES 17–16 Exchange vs 18–17 Local; @CL and @GC 18–17 either way); (3) sessions are per data series and multiple sessions per day are kept; (4) @6E priceScale is within the 1e-9 tolerance; (5) SPY is Stock / TickCount / 04:15–20:00; (6) `SessionName` returns the chart's template name. **NEW MC behaviour:** `LastBarInSession` did not fire at the GLOBEX 15:15 break, only at 16:00 (and on the 15:30–16:30 session it fired at 16:00, before the listed 16:30 end, because data stops at 16:00) → R14, MI-T16. **CLOSED 2026-11-04 (Ettienne, `5c9ccd2`): Portfolio Trader per-symbol test.** One PT log (Local tz) for @6E, @ES and AAPL; each stream printed its own root, tick, bpv, sessions and calendar:

- **ES:** tick 0.25, bpv 50, `cost(17.00 src=Ettienne 2026-10-31)`, `floor(14)`, `rth(09:30-16:15 src=EquityIndexLegacy)`, profileHash `e1927ccf1ea6f41e` (Local tz; reproduced on the box from the Default list `Sun–Thu 18:00→17:00`).
- **AAPL:** `class(Stock)`, `preset(none)`, tick 0.01, `calendar(Nyse)`, `rth(09:30-16:00 src=ChartDaySession)`, profileHash `41886799d411dd0d`.
- **@6E:** profileHash `9523b25025f9ed5b` (Local, sessions 18:00→17:00 ET) vs `55576df6a8a99c05` on the Exchange-tz chart: the evidence for D2 (§12.4).

The PR-M0 question (did @CL not start?) is moot. Ettienne's earlier @6E rerun is closed (below).

CLOSED (rev 2.3): @6E zone. Ettienne's QuoteManager had @6E on **Use Custom Sessions**, Sun–Thu 18:00 → next day 17:00, with Session TimeZone = **Exchange (CT)**: ET hours entered as CT. A user configuration error, not MC behaviour (CME 6E trades 17:00–16:00 CT like ES, research `/workspace/m-instr/mc-session-templates-research.md` "Observation: 6E", https://www.cmegroup.com/trading/why-futures/welcome-to-cme-fx-futures.html). He switched

@6E to the Exchange & ECN sessions and restarted MC; the rerun `01M6HTKV3SV1Q5QDXBFTCH8D3K.txt` reads Sun–Thu 17:00 → 16:00 Exchange, first bar 17:06, session ends 16:00, matching ES. It is the 6E fixture; `01M6HSDCCJZXJHNB165KSF6Z5B.txt` was misconfigured and is discarded (dropped from the fixture list).

4.4.3 `_RootDump` results (Ettienne `_RootDump` 2026-11-03, relayed by Coder; dxFeed)

Symbol	root	exch	cat	ccy	SymbolExpiry	minMove	PriceScale struct vs Info	BPV struct vs Info	UnsupportedFields
@ES	ES	CME	FUTURE	USD	2038-01-01	25	0.01 vs 100	50 vs 50	0
@CL	CL	NYMEX	FUTURE	USD	2038-01-01	1	0.01 vs 100	1000 vs 1000	0
@6E	6E	CME	FUTURE	USD	2038-01-01	5	1E-05 vs 99999.999999999985	125000 vs 125000	0
SPY	SPY	ARCA	eMTPA_MCSC_STOCK	USD	1900-01-01	1	0.01 vs 100	1 vs 1	0

`SymbolName` = `Info.Name` in every case. `MTPA_SymbolFields` values as in §4.1.

4.5 Sessions from the chart (R12; DECIDED Q1/Q2)

What MC does (research doc `/workspace/m-instr/mc-sessions-research.md`). Each data series has its own session setting (Format Instrument → Settings → Sessions = **Default**, i.e. the symbol's QuoteManager sessions, or a named template); in Portfolio Trader each symbol has its own session but the time zone is one global setting. Scripts cannot set sessions. Data outside sessions is filtered out; time bars always break on session boundaries and align to the session start; `Time` is the bar close. Holidays are removed by QuoteManager holiday lists and **cannot be read from code**. **Templates (research `/workspace/m-instr/mc-session-templates-research.md` §1, §3, §7):** a chart uses Default (the symbol's QuoteManager sessions: exchange sessions, custom sessions, a custom template or the symbol dictionary) or a named template; templates, built-in ones included, are editable in **QuoteManager → Tools → Session Templates** (Add/Edit/Delete, rows with Session Time and Session End; https://www.multicharts.com/trading-software/index.php?title=Setting_Custom_Session_Templates, https://www.multicharts.com/discussion/viewtopic.php?t=5890), but not from a script (staff, https://www.multicharts.com/discussion/viewtopic.php?t=53293). **"Combined" templates mean two sessions per day** (overnight + day, two daily bars; https://www.multicharts.com/discussion/viewtopic.php?t=45183), not one merged 23-hour session; "Open Outcry"/"Open Auction" templates describe pit sessions closed in 2015 and survive only as RTH-style windows.

Setup note (MC caches sessions). MultiCharts caches QuoteManager symbol sessions until a restart: after changing a symbol's sessions in QuoteManager, **restart MultiCharts** before running charts, dumps or tests (the @6E rerun, §4.4.2, needed it).

Environment caveat (not an engine rule). The built-in **"CME: Equity Index Futures [E-mini] (GLOBEX)"** template is stale: its 15:15–15:30 CT gap is the halt CME removed on 27/28 June 2021 (https://www.cmegroup.com/notices/electronic-trading/2021/06/20210621.html), and its 15:30–16:30 row matches the hours **before November 2012** (2012–2021 sessions ended 16:15, https://www.cmegroup.com/tools-information/lookups/advisories/market-data/20121112.html; pre-2012 sheet https://www.dormantrading.com/TraderTools/CMEeminisp500.pdf); current ES hours are 17:00–16:00 CT (https://www.cmegroup.com/markets/equities/sp/e-mini-sandp500.html). This template is on the **D9a/canary chart** (§4.4), so that chart has a session boundary at 15:15 and a listed end at 16:30. Engines **take sessions as given** (R12) and do not correct or warn about it; changing the canary chart's template is Ettienne's environment decision and would change MI-T15 baselines.

Reading. `SessionReader` (host) reads `BarsOfData(n).Sessions` into the MC-free struct `_MtfMath.Session` (#88 head, L47–52: `StartDay, StartTime, EndDay, EndTime`) **in StartCalc** (populated there, CONFIRMED §4.4.2) and **re-checks on the first CalcBar of each stream** (must be identical, else Error + throw `sessions:changed`, S1). Session ends follow R14 (from the list's `EndTime`s, not `LastBarInSession` alone). Null or empty sessions → throw `sessions:none` for every session-using engine (§6.4 S1). Template name: `Info.SessionName` on the INSTR line, log only. Whether the chart uses Default is not exposed (not documented) and is not checked: the chart's template is the truth.

Zone. Session times are in the bar time zone of the stream, i.e. the chart's Format Instrument time zone (**CONFIRMED** A3, §4.4.2: @ES 17:00–16:00 on Exchange, 18:00–17:00 on Local, bar times shift by the same hour; @CL/@GC identical either way). No conversion is needed before deriving. ET-anchored logic that remains (EVENT rows, the frozen G0/REGGATE trial windows) converts bar time to ET as today: Local → `TimeZoneInfo.Local`, GMT → UTC, Exchange → the `exchangeTimeZoneId` map (L2614–2626, kept only for this; its NYMEX/COMEX entry stays Eastern: A3 shows MC's exchange time for @CL/@GC equal to ET on Ettienne's setup, row 4).

Derived fields (MC-free, `<instr-core>`): - `tradingDayKey(t)` = the calendar date, in session time, of the **end** of the session that contains bar close `t` (start exclusive, end inclusive). Several sessions ending on the same date share the key (e.g. the GLOBEX template 17:00–15:15 + 15:30–16:30, `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`). On the CME Globex template (17:00–16:00 CT) this equals today's `GlobexTradingDayEt` (18:00 ET roll) for every bar; on a 24×5 FX template ending 17:00 ET it equals today's ET+7h rule; on a 00:00 UTC crypto template it equals the UTC date. A bar outside every session (should not occur, MC filters) → Error + throw `sessions:bar-outside`. - `dayOpen` = start of the first session of the trading day (TOD minute-of-session; replaces 18:00 ET at VPFUT L489–493 and the default at L949). - `sessionEnd` = end of the last session of the trading day, from the list (R14; on the GLOBEX template 16:30 even though the last bar closes at 16:00). **PR-M6b (Ettienne 2026-11-09 08:04 ET):** the FLOW tick-chunk boundary is that listed end converted to US Eastern, not the old fixed 17:00 ET. On the stale Globex template that is 17:30 ET. ES chunking on that template changes; the owner accepted it. - `hasOvernight` = some session of the trading day crosses midnight in session time (`StartDay != EndDay`). Globex 17:00 CT Sun → 16:00 CT Mon: true; an 08:30–15:15 day session: false. For 24-hour single-day templates (e.g. 00:00–24:00 crypto) the value is fixed in PR-M4 from the A2 output (**VERIFY**). - `rthSplit` = the RTH segment inside the trading day: **when the chart's sessions are day-only (`!hasOvernight`), the chart session is the RTH**. For a 24-hour chart (N1 DECIDED): ES/MES/NQ/MNQ use today's equity-index split from `_CmeSessionClock` (RTH 09:30, value end 16:00, session flag 16:15 ET), logged `rth src=EquityIndexLegacy`; any other root on a 24-hour chart has `rthSplit = null` (S7). No RTH window in any preset. - `sessionsPerWeek`/`sessionsPerYear`: annualisation stays a class constant (252 futures/stocks, 365 crypto, 260 FX **VERIFY**), not derived.

Multi-stream. Each stream has its own derived fields. The #86/#88 DV/MTF Warning when Data2..N sessions differ from Data1 (`checkHtfSessions`, #88 head L1802–1822) stays unchanged, including its MC-1439 / Realtime-History Matching caveat: with RHM on and different session ends, the strategy stalls after the shorter session's end until the auxiliary series prints. In Portfolio Trader, mismatched bar-close times across symbols can make the portfolio money-management signal run several times per bar (MC staff); M-PORT behaviour is unchanged, and the INSTR line shows each symbol's sessions so a mismatch is visible.

Removed because they existed only for hardcoded times (PR-M4 unless noted): `_CustomSignal` L1888–1889 (`"Eastern Standard Time"` + `GlobexTradingDayEt` for the roll key → `tradingDayKey`); `autoTradingDay` L2581–2587 + L2462 (→ `tradingDayKey`); VPFUT `GlobexDate` key L361 and TOD 18:00 L489–493; `_CmeSessionClock` Globex open/close/halt defaults L48–61 as trading-day inputs; `TickChunkBoundaryEt` (PR-M6b). The exchange-zone map stays only for ET conversions (above). Holiday calendars stay (they are not session times; §6.4 S4, PR-M8).

ES bit-identity. On the CME Globex template (D9a and canary charts, `_DvpEngineTest` L28–48, `_TestGate0` L42) the derived day key and day open equal today's constants. The listed session end on the stale template is 16:30 CT, which is 17:30 ET, not the old FLOW constant 17:00 ET (PR-M6b; Ettienne accepted that ES chunking change). On a chart with another template (e.g. Ettienne's live @ES Default sessions, listed end 16:00 CT = 17:00 ET) results follow that chart; this is the intended behaviour and is logged on INSTR, not warned.

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5. Data model

5.1 `InstrumentProfile` (immutable; MC-free core `<instr-core>` in `_InstrumentProfile.Function.CS`)

Field	Type	Source	Notes
`assetClass`	`EAssetClass { Future, Stock, SpotFx, Crypto, Index, Unsupported }`	detect (§4.2)	
`root`	string	`ASymbolInfo2.SymbolRoot` (CONFIRMED), parser only as fallback (§4.3 step 3)	"ES", "6E", "EURUSD", "BTCUSD", "AAPL"; never empty
`presetId`	string or null	preset	
`tick`	double	preset tick when a preset matched (after the §4.3 cross-check), else `MinMove/Info.PriceScale`	> 0 else `tick` refusal (no preset) / `mismatch:tick` (preset); printed with G15
`bigPointValue`	double	preset bpv when a preset matched, else `Info.BigPointValue`	currency per 1.0 price move (renamed from rev-1 `pointValue`; MC's `Info.PointValue` is another member)
`tickValue`	double	tick × bigPointValue	of the carried values (ES 12.5, 6E 6.25)
`currency`	ISO code	`ASymbolInfo2.SymbolCurrency` (CONFIRMED, §4.4 B2)	non-USD throws (Q8)
`sessions`	`_MtfMath.Session[]`	`Bars.Sessions` (§4.5) via `SessionReader` (IP L996–1006)	read at StartCalc, re-read and compared on the first bar (`WithSessions`, IP L814–822)
`sessionName`	string	`Info.SessionName`	log only
`barZone`	`RequestTimeZone` + resolved tz id	`Info.TimeZone` (+ map for Exchange)	ET conversions only
`tradingDay`	derived from `sessions`	§4.5	key = session-end date
`dayOpen`, `sessionEnd`, `hasOvernight`	derived	§4.5	
`rthSplit`	(start, end) or null, + `rthSource ∈ {ChartDaySession, EquityIndexLegacy, None}`	§4.5, N1	
`calendar`	`ECalendar { CmeEquityIndex, Nyse, NyseStandIn, Observed, None }` (IP L33; + CME product groups when verified, PR-M8)	preset; no preset: Stock → `Nyse`, Unsupported → `None`, else `Observed` (IP L252)	holidays/early closes only; no session times. `NyseStandIn` = NYSE business days (`EventRules.IsNyseBusinessDay`) standing in for the CME calendar for CL/GC/ZN/6E until PR-M8; explicit and logged (`calendar(NyseStandIn; calendar: NYSE stand-in (CME calendar pending M8))`)
`contractMonths`	month mask	preset	
`expiry`	`EExpiryRule` + params	preset; a dated contract's `SymbolExpiry` when strictly inside (1900-01-01, 2038-01-01) (§4.4 B1)	§8.2
`roll`	`ERollRule` + params	preset	§8.2
`sessionsPerYear`	int	class	252 / 365 / 260 (FX **VERIFY**)
`cost`	`CostModel { roundTripCents (long), source }`	`InstrumentConfig.costs`	Unknown when absent
`costFloorTicks`	int	§9 (integer cents); computed **only when a preset matched**, with the preset tick/bpv (IP L778–779)	-1 = Unknown (no preset, no cost, or invalid config); printed `floor(Unknown)`
`cluster`	string	preset	same strings as L1611–1614
`volumeBasis`, `volumeQuality`		`Info.BVO` + class	R7
`dataNotes`	string or null	`InstrumentConfig.dataNotes[root]`	Q7; printed as `dataNotes(…)` only when configured. **DIVERGENCE** from rev 2.3 "list of (from, what)": it is one free-text string per root

Field	Type	Source	Notes
`sizingUnit`	Contracts / Shares / Units	class	

5.2 Presets: exchange facts only (no times)

`InstrumentPresets`: static table in `<instr-core>`, one row per root (§8): tick, bigPointValue, currency, contractMonths, expiry, roll, calendar, cluster, source. Adding a root = a one-row PR with sources.

5.3 `InstrumentConfig` (protected on `_CustomSignal`; no inputs)

- `enabled` (when false: `buildProfiles()` skipped — no profiles, no refusals, no INSTR / profileHash; host tick from MinMove/PriceScale).
- `overrides`: typed `InstrumentPresetRow`, null by default (fix an exchange fact without a code PR; contains no session fields).
- `costs`: **asset class + root** → round trip in **cents** (`long`) (**D1 DECIDED**, Ettienne 2026-11-04, §12.4). One `costSource` string for all (default `Ettienne 2026-10-31`, printed `cost(17.00 src=…)`); values > 0.
  - **Key format:** `<Class>:<ROOT>`, where `<Class>` is the `EAssetClass` name exactly as `InstrumentText.ClassName` prints it (`Future`, `Stock`, `SpotFx`, `Crypto`, `Index`; `Unsupported` is not allowed) and `<ROOT>` is the resolved root, trimmed uppercase, as on the INSTR `root(…)` field. Code builds keys with one helper, `InstrumentConfig.CostKey(EAssetClass, string root)`, so no code parses a class name back from text; `Validate` accepts a key only when it equals `CostKey(c, r)` for some class `c ≠ Unsupported` and a non-empty uppercase `r`. A root-only key (`ES`) or a misspelt class is invalid → `unknown-field:config` (no silent migration of old keys).
  - **Defaults map:** ES 1700 → `Future:ES` 1700; MES 250 → `Future:MES` 250. `costSource` unchanged.
  - **Lookup:** `CostFor(assetClass, root)` reads only `CostKey(profile class, profile root)`.
  - **Class mismatch:** when no row exists for the symbol's own class but a row exists for the same root under another class (e.g. a stock `ES` while `Future:ES` is configured), that row is **not used**: the profile has no cost (`cost(Unknown)`), the floor is Unknown with its reason (`floor(Unknown src=cost-class-mismatch)`, §6.1/§9), and one Warning per stream at StartCalc names it: `INSTR ds(<n>) cost row Future:ES not used: symbol class Stock root ES`. Never silently applied, never silently dropped.
  - **Canon:** `InstrumentConfig.Canon()` keeps its form with the new keys, sorted ordinal: `costs=[Future:ES:1700,Future:MES:250,]` (cents after the last `:`). The profile Canon is unchanged (`cost=<cents>` or `-`; floor `-1` when Unknown; the src reason is printed, not hashed), so the pinned ES/MES/6E profileHashes do not change; only a profile that today shows a cross-class cost (stock `ES`: `cost=1700` → `cost=-`) gets a new hash.
  - **configHash:** not affected. On `5c9ccd2` `InstrumentConfig.Canon()` (IP L413–424) is called only by tests (IT L457, L546); it is not part of the RUN line (CS L2296–2302: `inputs[…]`, `event[EventConfig.Canon()]`, profileHash suffix) or of any engine `configHash` (e.g. `_DvpEngine` hashes `DvpConfig.Canon()` only). If a later PR (PR-M7 cost floor) puts `InstrumentConfig.Canon()` into a hash, it must be with the class+root keys from the start.
  - **M3 behaviour (as merged, superseded by PR-M4):** keyed by **root only** (IP L398, `CostFor(root)` L426–430; keys trimmed uppercase): a stock and a future with the same root share a cost (a stock ticker `ES` shows `cost(17.00 …)`, IT L609–611). M3 limits the damage by computing the floor only for a matched preset (§9).
- `applyCostFloor` (PR-M7, default false).
- `dataNotes`: root → note string.
- `enabled = false` skips `buildProfiles()` with one Info line `INSTR off (InstrumentConfig.enabled = false): no profiles, no cross-checks` (CS L2392–2396 @5c9ccd2): no INSTR lines, no RUN suffix.
- An invalid config (`Validate() != null`) is the log-only refusal `unknown-field:config`; its `overrides` row is not trusted for the cross-check (§6.2).
- `Validate()`, `Canon()` as in existing configs; `profileHash` outside every `configHash` (rev-3 Q7).

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6. Logging, application and refusal

6.1 Lines (`engineSink(level)`, one `File.AppendAllText` per line, InvariantCulture)

- `INSTR ds(<n>) name(<Info.Name>) cat(<Info.Category label>) class(<EAssetClass>) root(<r> rootSrc=mc|parsed) preset(<id>|none) exch(<Info.Exchange>) tz(<setting>→<zone id>|unresolved|unavailable(<ExceptionType>)) sessionName(<Info.SessionName>) sessions(n=<k> <Day hh:mm>→<Day hh:mm>; …) [xsessions(<exchange zone id> n=<k> …; rth <hh:mm-hh:mm>|none) — PR-M4, D2] dayKey(SessionEnd) dayOpen(hh:mm) sessionEnd(hh:mm) overnight(yes|no) rth(<start>-<end> src=EquityIndexLegacy|ChartDaySession) | rth(none src=None) tick(<G15>) bpv(<G15>) tickValue(<G15>) ccy(USD) bvo(<label>) calendar(<ECalendar>[; <note>]) months(<letters>) expiry(<rule>|none) roll(<rule>[(n)]|none) cost(<c.cc> src=<costSource>)|cost(Unknown) floor(<n>)|floor(Unknown) cluster(<c>|none)[dataNotes(<note>)] profileHash(<16 hex>)` at **Info**, once per stream after the first-bar session read (`InstrumentProfile.InstrLine`, IP L846–872 @5c9ccd2). `dataNotes(…)` appears only when `InstrumentConfig.dataNotes` has the root; it sits after `cluster` and before `profileHash`. `cat`, `bvo` and the tz setting are MC enum labels formatted by the host for the log only (R1; `unavailable(<ExceptionType>)` when the typed read throws, CS L2449–2474). The tz zone part is the resolved `TimeZoneInfo.Id`, `unresolved` when no zone resolves, or `unavailable(<ExceptionType>)` (CS `instrZoneLabel` L2465–2471). **Pinned (rev 2.4; the tests pin these exact strings).** ES, CME GLOBEX 9-entry list (IT T12 L537–539): `INSTR ds(1) name(@ES) cat(Future) class(Future) root(ES rootSrc=mc) preset(ES) exch(CME) tz(Exchange→Central Standard Time) sessionName(test) sessions(n=9 Sun 17:00→Mon 15:15; Mon 15:30→Mon 16:30; Mon 17:00→Tue 15:15; Tue 15:30→Tue 16:30; Tue 17:00→Wed 15:15; Wed 15:30→Wed 16:30; Wed 17:00→Thu 15:15; Thu 15:30→Thu 16:30; Thu 17:00→Fri 15:15) dayKey(SessionEnd) dayOpen(17:00) sessionEnd(16:30) overnight(yes) rth(09:30-16:15 src=EquityIndexLegacy) tick(0.25) bpv(50) tickValue(12.5) ccy(USD) bvo(Volume) calendar(CmeEquityIndex) months(HMUZ) expiry(ThirdFridaySoq) roll(EquityIndexCustomary) cost(17.00 src=Ettienne 2026-10-31) floor(14) cluster(EQIDX) profileHash(020deca78d6392b7)` 6E, same template, NyseStandIn calendar (IT T13 L475; tz label as the test passes it): `INSTR ds(1) name(@6E) cat(Future) class(Future) root(6E rootSrc=mc) preset(6E) exch(CME) tz(Exchange) sessionName(test) sessions(n=9 Sun 17:00→Mon 15:15; … same 9 entries as ES…) dayKey(SessionEnd) dayOpen(17:00) sessionEnd(16:30) overnight(yes) rth(none src=None) tick(5E-05) bpv(125000) tickValue(6.25) ccy(USD) bvo(Volume) calendar(NyseStandIn; calendar: NYSE stand-in (CME calendar pending M8)) months(HMUZ) expiry(ThirdWedMinus2Bd) roll(BusinessDaysBeforeExpiry(5)) cost(Unknown) floor(Unknown) cluster(FX_EUR) profileHash(8e8b3e9b0e230e12)` The 6E hash `8e8b3e9b0e230e12` is for the **synthetic** 9-session GLOBEX list the test passes. **MC result (MI-T15, Ettienne 2026-11-04):** the real @6E chart (Exchange tz, Default session, BVO TickCount) printed `tick(5E-05) bpv(125000) tickValue(6.25)`, the NyseStandIn calendar note, `roll(BusinessDaysBeforeExpiry(5))` and profileHash **`55576df6a8a99c05`**; the box (.NET 8) reproduces it byte-for-byte from the canon with `sessions=n=5 Sun 17:00→Mon 16:00; … Thu 17:00→Fri 16:00`, so the hash is **runtime-stable** (.NET Framework = .NET 8). It differs from the pin only by the session list. **Rev 2.5.2.** `profileHash` is FNV-1a 64 of `InstrumentProfile.Canon()` (`_InstrumentProfile.Function.CS` L1013–1031): class, root, preset, tick, big point value, currency, calendar, months, expiry, roll, cost cents, floor, cluster, `sessionsPerYear`, and the session list plus RTH split **in exchange time**. The ES pin `020deca78d6392b7` is that canon for the ES preset on the 9-entry GLOBEX list (`tests/InstrumentTests/Program.cs` `Globex9()` L521–524, the `01M6HSDBFRQ9ZZFA` dump; pin check L712) with bar zone equal to the exchange zone (Central). That is the same list the spec already says sits on the D9a/ canary chart (the stale "CME: Equity Index Futures [E-mini] (GLOBEX)" template, §4.5 environment caveat). A native Exchange-tz CME chart of that template therefore prints `020deca78d6392b7`. The Nov 8 D9a runs did, on both streams. That is consistent. It is not the Default-session pin `6cff053d1b215324`. The 2026-11-04 note that this pin was "not expected in MC" (§12.3) is withdrawn for the ES 9-session chart. `8e8b3e9b0e230e12` is still the 6E row of the same list; the Nov 8 run was @ES and did not print it.

- **Number formats.** Profile facts on the INSTR line and in the hash canon use `InstrumentText.Fact` = `ToString("G15", InvariantCulture)` (IP L111): `5E-05`, `0.25`, `12.5`. `InstrumentText.Num` = `"R"` (IP L108) is used only inside diagnostics (mismatch and refusal texts), e.g. the raw 6E tick `5.0000000000000009E-05` (.NET Framework; .NET 8 prints `5.000000000000001E-05`). Diagnostic digits may differ between runtimes; profile text and hash may not.
- **profileHash** = FNV-1a 64 over the canonical profile text (`Canon`, IP L825–835; `ProfileHash` L836–841): class, root, preset, tick, bpv, ccy, calendar, months, expiry, roll, cost cents, floor (-1 when Unknown), cluster, sessionsPerYear, the session list and the RTH split (**in exchange time**, D2, as merged in PR-M4). Not hashed: name, exchange, sessionName, the tz/cat/bvo labels, costSource, `dataNotes`, tickValue and the derived day fields. **RUN suffix (M4):** the RUN line ends with `profileHash(<ds>=<hash>;…)` over the **started** streams only (a stream counts as started only once `Build` returned a profile; every refusal throws, so there is no `none` entry any more); empty when M-INSTR is off or no stream started (`InstrumentStreams.RunSuffix`, IP L953–963; appended at CS L2302 after the `configHash` part). It is **outside every `configHash`** (rev-3 Q7); IT T12 L544 pins the format.
- **Log-only line (M3 only; removed in M4):** at **Warning**, `INSTR ds(<n>) log-only, not enforced until PR-M4: <refusal>`. Under M4 every refusal is Error + throw; the log-only helper and the null-profile path are gone. Failed typed / session reads also Error + throw (`instrReadFailed`, CS).
- **Root cross-check** (§4.3 step 3), once per stream at StartCalc, **Warning**: `INSTR root-crosscheck mc(<r>) parsed(<p>) name(<n>)`.
- **Unexpected exception inside a guarded step**: `INSTR ds(<n>) unexpected <ExceptionType>: <message>` at **Error**, then rethrown (IP L925).
- **M-INSTR off**: `INSTR off (InstrumentConfig.enabled = false): no profiles, no cross-checks` at Info (CS L2394); no INSTR lines, no RUN suffix.
- **D2 (DECIDED, Ettienne 2026-11-04; PR-M4): profileHash over sessions in exchange time.** Today `Canon` hashes `InstrumentText.Sessions(sessions)` in the **bar** time zone, so one instrument hashes differently per chart/PT zone: @6E Exchange-tz chart (Sun 17:00→Mon 16:00 … Central) `55576df6a8a99c05`, @6E in Portfolio Trader on Local (18:00→17:00 ET) `9523b25025f9ed5b`. From PR-M4:
  - **Zones (as merged, PR-M4 @d3c0c1a).** Bar zone = `strictBarTimeZone(Bars.Info)` (CS L2954+): Local → Local, GMT → Utc, Exchange → the strict exchange lookup, anything else → null (no Local fallback). Exchange zone = `exchangeTimeZoneId(Info.Exchange)` (virtual) → static `exchangeZoneIdOf` (CS L2587–2615, `<instr-host-core>`): split the name on non-alphanumerics, exact case-insensitive token match, **first matching token wins**. Chicago → `Central Standard Time`: CME, XCME, GLOBEX, CBOT, ECBOT, CBT, XCBT, CBOE. New York → `Eastern Standard Time`: NYMEX, COMEX, NYSE, NASDAQ, ARCA. Looked up **whatever the chart's `Info.TimeZone` setting is**. The old `findTz` Local/`America/New_York` fallbacks are not on the profile path.
  - **Refusal, never a silent fallback.** Bar zone unresolved → `tz:bar`; `exchangeZoneIdOf` returns null or the id does not resolve → `tz:exchange` (§6.3; Build refuses and throws — there is no `profileHash(…=none)` path under M4). Action text: "add the exchange to exchangeZoneIdOf" / "set the chart time zone". A symbol on an exchange outside the map (FX, crypto feeds, and composite names such as NYSEARCA / NASDAQGS / CMEGLOBEX / AMEX / BATS / CFE) is refused until its exchange is mapped (**A3 OPEN**, §12.4).
  - **Conversion.** Each session's start and end (`DayOfWeek` + time) is converted bar zone → exchange zone on the trading week of the first bar (Sunday on or before the first bar's date, bar time), via `TimeZoneInfo.ConvertTime`; identity when the two zones have the same id. Same-DST zone pairs (ET↔CT) give the same text in any week. **DST coverage in IT (M4):** summer week, spring-forward gap, and fall-back ambiguous hour (read as standard) all keep 6E Default pin `55576df6a8a99c05`. A **GMT** chart remains **VERIFY** (does MC shift GMT sessions across DST?); one MC GMT run is still open (§12.4 A3).
  - **What is hashed.** `sessions=` in exchange time, and `rth=` recomputed by `RthSplit.Of` on the exchange-time list (rth **is** in `Canon`, IP L834; a `ChartDaySession` rth is session-derived; `EquityIndexLegacy` is fixed ET constants and is unchanged). The zone id itself is **not** hashed (Windows `Central Standard Time` vs `America/Chicago` on .NET 8 would break runtime stability). `dayOpen`/`sessionEnd` are not hashed and stay in bar time.
  - **What engines and the INSTR line use.** The profile keeps `sessions`, `rthSplit`, `dayOpen`, `sessionEnd` in **bar time** for the engines (R12, bar times are in bar zone). The INSTR line keeps `sessions(…)`, `dayOpen`, `sessionEnd`, `rth(…)` in bar time and adds `xsessions(<exchange zone id> n=<k> …; rth <hh:mm-hh:mm>|none)` = exactly the hashed exchange-time text.
  - **Pins as merged (box + IT; confirm equal hashes in MC on Exchange chart vs PT Local).** @6E Default `55576df6a8a99c05` on **any** chart/PT zone (PT Local `9523b25025f9ed5b` goes away); @ES Default (Sun–Thu 17:00→16:00 CT, EquityIndexLegacy rth) `6cff053d1b215324` on any zone (PT Local `e1927ccf1ea6f41e` goes away); AAPL `41886799d411dd0d` unchanged when the PT PC is on ET (bar zone = exchange zone). The IT pins ES `020deca78d6392b7` / 6E `8e8b3e9b0e230e12` stay **only if** the test declares bar zone = exchange zone; every test pin built from a non-exchange bar zone changes, and the tests gain the invariance checks of §11.1 item 9.
- **Floor-Unknown reason (DONE in PR-M4):** `floor(Unknown src=no-preset|no-cost|cost-class-mismatch|config-invalid)` (§9, `FloorSource`).
- `CFG <ENGINE> applied{…} undefined{output=reason;…} degraded{mode}` at Info, once per engine.
- `SKIP <ENGINE> reason(<code>) …` at **Info**.
- Mismatches (and every other enforced refusal, §6.2) go out at **Error** before the throw.

6.2 Application

`buildProfiles()` runs in `StartCalc` for instrument fields; session-derived fields are filled on the first CalcBar of each stream, and engines whose configuration needs them (TOD, AVWAP session date, FLOW chunk plan, RANGE/WEEKLY session sources, VPFUT day key) finish configuring there, before their first update. Engines get the profile through their existing config objects; engines never read `Bars.Info` or `Bars.Sessions` directly.

As merged in PR-M4 (#98, @d3c0c1a); every refusal enforced globally (A1). Engines still do not read a profile yet (they move on from PR-M5).

- `buildProfiles()` (CS L2406+) runs in StartCalc **before** the engines start. Per stream: typed read (`InstrumentMcFields.Read`), `InstrumentProfile.Build` via `InstrumentStreams.Start` / `Guard`, StartCalc sessions via `SessionReader`. `profilesFirstBar()` runs on the first CalcBar, before the RUN line: re-reads sessions, `InstrumentSessions.Require` (S1), derives day key/open/end/RTH split (`WithSessions`) and prints the INSTR line.
- **`Build` is two-phase** (IP L858–922 @d3c0c1a). **Phase 1, the cross-checks**, runs only when the class is not Unsupported: root pick → `Checks.McInternal` (`mismatch:mcPriceScale` / `mcBigPointValue`) → raw tick → preset choice (valid `overrides` row matching class + root, else `InstrumentPresets.Find`) → `Checks.Preset` when a preset matched (`mismatch:tick` / `mismatch:bigPointValue`) → the cost floor when the config is valid and a preset matched (`mismatch:tickValue`, §9). **Phase 2, the refusals** (all enforced): `class` → `unknown-field:config` (invalid `InstrumentConfig`, incl. old root-only cost keys) → `unknown-field:root` → `tick` (PriceScale ≤ 0 / tick ≤ 0, no preset) → `currency` → `calendar` (when the profile has an expiry or roll rule, R6) → `tz:exchange` / `tz:bar`.
- **Enforcement (M4, A1).** `InstrumentStreams.Guard` buffers the Error line; **every** `InstrumentException` writes it and rethrows (the study stops). There is no log-only path and no null-profile / `profileHash(…=none)` path. Any other exception writes `INSTR ds(n) unexpected …` at Error and rethrows. **`Start` sets `m_started` / `m_pending` only after `Guard(Build)` returns** — a refused stream is never marked started (`InstrumentStreams.Start`, IP).
- **Session refusals on the first bar** (`sessions:none`, `sessions:changed`, `sessions:bar-outside`, …) are likewise Error + throw under M4 (`FirstBar` → `Guard(WithSessions)`).
- **What is refused vs what still runs (confirmed @d3c0c1a).** Refused (Error + throw): Unsupported / unmapped class; non-USD; unresolved bar or exchange zone (incl. unmapped exchange → `tz:exchange`); PriceScale ≤ 0 / tick ≤ 0; missing or changed sessions; failed typed or session reads; invalid `InstrumentConfig` (incl. old root-only cost keys → `unknown-field:config`); `mismatch:*`; `calendar` / `months` when rules need them. **Still runs:** futures with no preset (cost / floor Unknown, `src=no-preset` or `no-cost`); USD stocks on NYSE / NASDAQ / ARCA (mapped → Eastern).

6.3 Exception format

`M-INSTR: <ENGINE> cannot run on <name> (class <c>, root <r>, sessions <sessionName>): <reason>. <action>`. Reason codes (`InstrumentReason`, IP L46–56 @5c9ccd2): `class`, `unknown-field:<f>` (`root`, `config`, `cost`), `tick`, `mismatch:<f>` (`tick`, `bigPointValue`, `mcPriceScale`, `mcBigPointValue`, `tickValue`), `volume-basis`, `currency`, `sessions:none`, `sessions:changed`, `sessions:bar-outside`, `sessions:no-rth-split`, `sessions:window-not-covered`, `calendar`, **`months`** (rev 2.4: `ContractFor` finds no

listed contract month within 24 months, action "check the preset's contract months", IP `ContractFor` L519–530; tested IT T4 L241–242), and from PR-M4 (D2) **`tz:bar`** (the bar time zone cannot be resolved) and **`tz:exchange`** (no exchange zone for `Info.Exchange`, or the zone id does not resolve). Under M4 every reason above is enforced (§6.2, A1).

6.4 R13 applied: every throw, degraded mode and undefined output

Throw rules | # | Condition | Engines | Action text | |---|---|---|---| | S1 | `Bars.Sessions` null/empty at StartCalc or on the first bar; or StartCalc and first-bar reads differ; or a bar outside every session | every session-using engine (TOD, AVWAP, VPFUT, FLOW, RANGE, WEEKLY, DAILY, DVP, G0, REGGATE, ROLL keying) | "check the chart's session template" | | S2 | An engine whose purpose needs an overnight segment, on a chart with `!hasOvernight` | none on main (every overnight-specific item is an output, see U1–U2) | — | | S3 | A frozen pre-registered window (G0 09:30–16:00 ET, `_Gate0` L474/L614/L1037…; REGGATE 09:30–16:00 ET, `_RegimeGate` L45–48) is not inside the chart's sessions (converted to ET) | G0, REGGATE | "use a session template that covers 09:30–16:00 ET" | | S4 | An engine needs a holiday or early close **in advance** and the root has no verified calendar (rev 2.4: `NyseStandIn` for CL/GC/ZN/6E is a stand-in, not verified; it serves the expiry/roll business-day rules only. A preset with roll/expiry rules and no calendar is refused `calendar` at Build, R6; enforced since M4) | REGGATE (early-close exit), G0 (early flat), VPFUT `HaltEt`/`RthCloseEt`/`IsGlobexClosedDay` hooks (L1635–1671), `_CmeSessionClock` holiday use | "add a verified calendar for (PR-M8)" | | S5 | A required preset field is Unknown (`roll`, `contractMonths`, `expiry`, `cost`) | ROLL Table/SeriesWindow, explicit EVENT own-expiry queries (not Step/Classify; 2026-11-06), DVP roll table, COST/cost floor | "add a preset row / cost for " | | S6 | Class Unsupported in the rev-3 matrix | per matrix | rev-3 reason text | | S7 | An engine needs `rthSplit` and it is null (24-hour chart, root not ES/MES/NQ/MNQ; N1) | VPFUT RTH profile/value window/IB, AVWAP `rth`, WEEKLY `RthSessions`, DVP RTH features | "use a day-session template for " | | S8 | tick ≤ 0; detected ≠ preset; futures on Tick Count (R7); non-USD (Q8) | all | — |

Approved degraded modes (only these; each from a DECIDED answer): D1 VP on FX/CFD/crypto as visual/level only, nothing feeds Holm (rev-3 Q5); D2 G0 on Index price-only parts (rev-3 matrix note 6; G0 is ES-canary only in practice); D3 EVENT on USD FX pairs, US events as a filter (rev-3 Q11); D4 FLOW on crypto, single venue, Warning (rev-3 reason 12); D5 PriceAction on Bond/MF/Unknown (rev-3 reason 2); D6 FLOW bid/ask features undefined before a root's `dataNotes` bid/ask start date (Q7, accepted 2026-11-02); D7 TOD on a root without a verified calendar: no current-day half-day flag, observed short days excluded from baselines, one Info line (N4 DECIDED). Rev 1's "Degrade" on RTH-only charts, on unknown early closes and on unconfirmed RTH windows is **withdrawn**. A new degraded mode needs Ettienne's approval.

Undefined outputs (NaN / false, one Info line, not a degrade) | # | Output | When | |---|---|---| | U1 | VPFUT overnight stats `ethHigh/ethLow/ethVolume` (L378) | `!hasOvernight` (N3 DECIDED) | | U2 | VPFUT `IsAsia()/IsEurope()` helpers (L244–245) | `!hasOvernight` (N3 DECIDED) | | U3 | DVP `rthShare` | `rthSplit == null` (already NaN on the generic base) |

Post-hoc calendar use without a verified table (correct, not degraded): `tradingDayCount` (L3228–3240) counts observed trading days (days that have bars) instead of NYSE days for roots without a table; `calendar = Observed`.

Own-contract availability (Ettienne 2026-11-06 12:13 ET, not a degraded mode and not a false result). When profile enforcement is on and the root has no roll facts, own-contract expiry/roll is `Unavailable`: `Step`/`Classify` still run, selected Macro rows and selected shared calendar rows (opex, VIX, rebalance, half day, Russell) still appear, and `isRollWeek` / `isQuarterlyExpiry` are null. They are not the equity-index dates and they are not false. `EventConfig.ownContract` off is `NotSelected` (also null, not false) and does not remove those shared rows. An explicit own-expiry/roll query throws S5. `NyseStandIn` does not become a verified calendar for `SessionsToExpiry`.

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7. Per-engine migration

Engine	Change	ES result (CME Globex template)	PR
CS host	L2462 typed; `autoTradingDay` and L1888–1889 → `tradingDayKey`; tick fallback → throw; `_NyseHolidays` for stocks, `p.calendar` otherwise; exchange map kept for ET conversions	identical	M1b, M4
CLOCK / VPFUT	Day key, day open, Globex open/close come from session data; RTH split per N1 (ES: today's constants, `rthSource=EquityIndexLegacy`); value window (09:30, 16:00] unchanged for ES; Asia/EU/NY windows per N5; holiday hooks per S4	identical	M4
CAL	ES keeps `_CmeHolidays` (both `UseCmeHolidayTable` flags, VPFUT L1627 and clock L81); NYSE for stocks; other roots `Observed` for post-hoc use and S4 throws for in-advance use until a verified table lands	identical	M8
ROLL	`_RollDetector` takes months/expiry/roll from the profile; F2; `gapTicks` stays 8 (Q4)	identical	M5
EVENT	Own `RollDate`/`ExpiryDate`/`SessionsToExpiry` from the profile when roll facts exist; separate `ownContract` / `sharedCalendar` controls (no inferred class mask). No facts: Step/Classify `Unavailable` (null, not false, not the ES quarter); explicit queries throw S5. NyseStandIn `SessionsToExpiry` still S4	identical	M5
G0	Remove `tick = 0.25` (L357; throw if 0) **and update all 35 `new Gate0Engine` constructions in `tests/Gate0Tests/Program.cs` to set `tick = 0.25`** in the same PR (none sets it today); `"ES"` (L1204) → `p.root`; S3 coverage check; windows frozen	canary hash identical	M6a
FLOW	Tick-chunk boundary = the template's listed `sessionEnd` converted to ET. Stale CME Globex: 16:30 CT = **17:30 ET** (was 17:00 ET). Spot FX stays Unsupported (rev-3 reason 11). 6E and other FX futures use their own listed end	**changes** on the stale Globex template (owner accepted 2026-11-09)	M6b
DAILY / REGIME	`hvAnnualisation` and the default `{252,126,63}` lookbacks read `p.sessionsPerYear` (ES/MES stay 252, 126, 63). REGIME `Tsmom63z` uses `sqrt(63/sessionsPerYear)`; the 63-session window stays	identical	M6c
LEVEL / AVWAP	No implicit 0.25 (`_LevelStudy` L80, `_AvwapManager` L154). Hosts set `tickSize()` (CS L1477, L2147). Gate 0 copies its tick in `Configure` (L471). AvwapTests helper sets 0.25 (L17). Avwap L196 rejects a tick that is not > 0. Level study does not throw	identical	M6d
RANGE / WEEKLY / PSTATS	`ringBars` literals kept (L31 2880, L47 1440). WEEKLY `RthSessions` with a filled profile and no `rthSplit` is S7; `GlobexSessions` is the user's other source. `isRthBar` stays. PSTATS 4-tick disagreement stays, with one CFG Info line when enabled	identical	M6e
PORT	`PortfolioCluster` returns `p.cluster`, same strings	identical	M6f
DVP	root from `p.root` (F1 done in M1a); roll table from the profile replaces `root == "ES"` (L213/230) and host loops (L119–121)	identical (post-#88 baseline)	M6g
REGGATE / `_Test6`	ES only (N2); windows frozen + S3; `tradingDay` hash text from an explicit `switch (ETradingDayKey) { case Globex18ET: return "Globex18ET"; … }` and `RollDetection` via an explicit switch with the same spellings (L62, `_Test6` L105–106); roll table from the profile (`_Test6` L100–102)	hash identical	M6h

Engine	Change	ES result (CME Globex template)	PR
SIZE / COST	`unitNotional` = entry × `p.bigPointValue`; cost floor = **new opt-in behaviour** (§9)	identical (off by default)	M7
MTF	`SymbolRoot` resolves via §4.3; doc fix (L28)	identical for ES	M1a
LEGACY	F3a/F3b only; others a one-line note	—	M1c

8. Presets: exchange facts only, no session times (sources read 2026-11-02)

8.1 Futures

Root	Tick	Big point value ($/1.0)	Tick $	Months	Expiry rule	Roll rule (Q5 DECIDED method)	Calendar	Cluster	Source
ES	0.25	50	12.50	HMUZ	3rd Fri 09:30 ET SOQ, back to the NYSE business day	Mon before the 3rd Fri (3rd Fri ≥ 2022-04-01), else Thu 8 days before (`_EventCalendar` L299–303)	CmeEquityIndex (`_CmeHolidays`)	EQIDX	CME ES specs; code
MES	0.25	5	1.25	HMUZ	as ES	as ES	CmeEquityIndex	EQIDX	CME Micro E-mini FAQ
NQ	0.25	20	5.00	HMUZ	as ES	as ES (**VERIFY** against the vendor splice)	CmeEquityIndex	EQIDX	CME Micro FAQ
MNQ	0.25	2	0.50	HMUZ	as ES	as NQ	CmeEquityIndex	EQIDX	CME Micro FAQ
CL	0.01	1000	10.00	all 12	3 business days before the 25th of the prior month (4 if the 25th is not a business day)	`BusinessDaysBeforeExpiry(2)`: 2 business days before LTD (**VERIFY** vs vendor splice)	NyseStandIn (logged; CME calendar PR-M8)	ENERGY	CME CL specs; CME notice chadv25-391 (CLF26 LTD 2025-12-19)
GC	0.10	100	10.00	active G J M Q V Z	third-last business day of the contract month	`BeforeFirstNotice(1)`: business day before first notice = last business day of the prior month (**VERIFY**)	NyseStandIn (logged; CME calendar PR-M8)	METALS	CME GC specs; GC fact card
ZN	0.015625	1000	15.625	HMUZ	7 business days before the last business day of the month	`BeforeFirstNotice(1)` (**VERIFY**)	NyseStandIn (logged; CME calendar PR-M8)	RATES	CME ZN specs; Barchart (ZNZ25 LTD 2025-12-19, FND 2025-11-28)
6E	0.00005	125,000	6.25	HMUZ (serials listed)	2nd business day before the 3rd Wed	`BusinessDaysBeforeExpiry(5)`: 5 business days (about one week) before expiry (**Ettienne 2026-11-04**, #96; replaces "Mon of the expiry week")	NyseStandIn (logged; CME calendar PR-M8)	FX_EUR	CME 6E specs; CME rule 26101.G

Notes. No session, RTH, halt or Globex-time column: those come from the chart (R12). Rev 1's RTH reference windows (CL 09:00–14:30, GC 08:20–13:30, ZN/6E 08:20–15:00 ET) and the ES 15:15–15:30 CT halt question are **dropped** (Q2 moot). `BigPointValue` for 6E is **CONFIRMED** 125000 (`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`); ZN is still **VERIFY** (not in the 13 runs); the cross-check (§4.3 step 8) refuses a mismatch. "Observed" calendars serve post-hoc use only; in-advance use throws (S4) until PR-M8 adds a sourced table. **Rev 2.4 (as merged, IP `InstrumentPresets` L215–237 @5c9ccd2).** CL/GC/ZN/6E use `ECalendar.NyseStandIn`: their expiry/roll rules count NYSE business days (`EventRules.IsNyseBusinessDay`, IP `InstrumentCalendars.BusinessDay` L439–446) until PR-M8 adds the CME calendars; the choice is explicit and the INSTR line says so (`calendar(NyseStandIn; calendar: NYSE stand-in (CME calendar pending M8))`, note text IP L437). NYSE and CME holiday schedules are not identical, so a computed date next to a holiday is VERIFY until PR-M8. No weekday fallback (R6). **MI-T4 fixture** `tests/InstrumentTests/fixtures/expiry-worked.csv` (11 rows: CL×4, GC×2, ZN×2, 6E×3): the `expiry` column was checked by the analyst on 2026-11-04 against CME notice chadv25-391, Barchart, CME rule 26101.G and the GC fact card; the `roll` column is **computed**: the preset rule applied by hand (e.g. 6EM24 expiry 2024-06-17, roll 2024-06-10), not vendor roll dates. The vendor-splice comparison stays VERIFY (PR-M5/M9).

8.2 Rule kinds (typed enums)

`EExpiryRule { ThirdFridaySoq, CL25thMinus3Bd, ThirdLastBd, LastBdMinus7Bd, ThirdWedMinus2Bd }`, `ERollRule { EquityIndexCustomary, BusinessDaysBeforeExpiry(n), BeforeFirstNotice(n), None }` (n is the row's `rollDays`; the INSTR line prints `roll(BusinessDaysBeforeExpiry(5))`). `ECalendar { CmeEquityIndex, Nyse, NyseStandIn, Observed, None }`: `CmeEquityIndex`, `Nyse` and `NyseStandIn` give a business-day function, `Observed` and `None` give none, so a rule needing one refuses `calendar` (R6). ES/MES/NQ/MNQ use `ThirdFridaySoq` + `EquityIndexCustomary` = today's `EventRules.RuleExpiryDate`/`RollDate`.

8.3 Non-futures

Class	Trading day	Calendar	Cost	Notes
Spot FX (12)	from the chart's sessions (a 17:00 ET-ending template gives today's ET+7h day)	Observed	Unknown (not costed until traded)	TickCount volume allowed (R7); DVP, G0, FLOW Unsupported (rev 3)
Crypto (20)	from the chart's sessions	Observed	Unknown	`sessionsPerYear` 365
Stock (2)	from the chart's sessions	Nyse (`_NyseHolidays`)	Unknown	FLOW and VPFUT Unsupported (rev 3). SPY (**CONFIRMED** `01M6HSDC62GCMK5ZANKB61PD5T.txt`): `Stock`, ARCA, Build Volume On = TickCount (tagged `TickCount`, R7, no throw), Default session Mon–Fri 04:15–20:00 Exchange time. That template is day-only, so by N1 its `rthSplit` is the whole 04:15–20:00 chart session, extended hours included; an RTH-only study needs a 09:30–16:00 template on the chart.

9. Sizing and costs (R8, R9): as the code is on `7dbdb1b`

1. q = 1% risk on the structural stop (CS-4 `sizeForRisk`, L2921), then `maxQty` (L2943) and the notional cap (L2945–2949) **reduce** q (Info when they bind). q never grows.
2. `willArm = m_portOn && bars == Bars` (L2957). - **Not willArm** (M-PORT off, or Data2..N): if q × stop × bigPointValue > 1% × equity the entry is **skipped**, Info "trade skipped" (L2958–2968). - **willArm** (portfolio Data1): the order goes to `portfolioArm` (L2970) and `_PortMM` applies deny-not-reduce. M-INSTR changes nothing here.
3. **Cost floor = new opt-in behaviour (PR-M7)**: `InstrumentConfig.applyCostFloor = false` by default (no input). When on, the stop is widened to `max(structural, costFloorTicks × tick)` before step 2, so the existing skip/deny paths act on the widened stop. Rev-1's claim that main already does this is withdrawn (`costFloorTicks` exists only in comments, `_Gate0` L1185, `_TestGate0` L145).
4. `costFloorTicks` in integer cents: `tickValueCents = tick × bigPointValue × 100`, which must be within 1e-6 of an integer (else throw `mismatch:tickValue`); `costFloorTicks = ceil(roundTripCents × 10 / tickValueCents)` in `long` arithmetic, `(a + b − 1) / b`. ES: 1700 × 10 / 1250 → 14. MES: 250 × 10 / 125 = 20 exactly (no float path can give 21). 4a. **As merged (M2/M3, IP `CostModel.FloorTicks` L379–389, Build L778–779 @5c9ccd2).** The floor is computed **only when a preset matched and the config is valid**, using the preset's tick and bigPointValue; otherwise `costFloorTicks = -1` (`floor(Unknown)`). A known cost with a negative cents value refuses `unknown-field:cost`; the non-integer tickValue is `mismatch:tickValue` (enforced in M3). Rationale: with M3's root-only keys (§5.3), a stock ticker `ES` must not get a floor from the ES future's cost. **PR-M4 (D1 DECIDED):** costs are keyed by class + root (§5.3), so the cost itself can no longer cross classes; the preset gate stays. PR-M4 also prints why the floor is Unknown: `floor(Unknown src=no-preset|no-cost|cost-class-mismatch|config-invalid)` (§6.1); `cost-class-mismatch` = a row exists for the root under another class and was not used.
5. Costs: `p.cost` from `InstrumentConfig.costs` (cents), keyed by **asset class + root** from PR-M4 (§5.3, D1 DECIDED 2026-11-04; M3 keys by root only). DECIDED: `Future:ES` 1700, `Future:MES` 250. Other roots: the method is DECIDED (Q3, IBKR fee per side ×2 + one tick slippage per side ×2), but the per-root IBKR fees are a **data input from Ettienne**; until given, cost = Unknown and cost users throw (S5).

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10. Tests (MC-free; new `tests/InstrumentTests` in `PLStudies.Tests.sln`, `CheckForOverflowUnderflow=true`, real sources linked via the `gen.sh` region pattern)

ID	Test
MI-T1	(Not a test.) `ClassOf` is the exhaustive host switch of §4.2, checked by the PLEditor compile in PR-M3. The MC-free core is tested on `EAssetClass` only. **Result 2026-11-04 (Ettienne, `5c9ccd2`): Compile All passed** after the one-time import of `_InstrumentProfile.Function.CS`; `eMTPA_SF_Root`, `eMTPA_MCSC_USD` compile.
MI-T2	Root resolver (§4.3): @ES, @ES#C, ESZ24, ESZ4, **ESZ2024 → ES, 'ES 12-24' → ES, ESZ24:XCME → ES, CLZ2026 → CL, @6E → 6E, 6EZ26 → 6E, EUR/USD → EURUSD, BRK.B → BRK (M1a; class-aware stock handling in the M2 resolver keeps BRK.B), Z4 → Z**, MESZ26 → MES, MESH27 → MES, **6E vs 6B differ**, @MNQ, GCQ26, ZNH27, **EUR/USD vs EUR/JPY differ**, BTCUSD, AAPL; M2: `/` removed for Spot FX only; M2 root source: MC root primary, parser fallback only on `(UnsupportedFields & eMTPA_SF_Root) != 0` (incl. -1) or empty root, `!SameRoot` → Warning + MC wins; **`_RootDump` fixtures (§4.4.3):** (@ES, ES, 0) → ES/mc, (@CL, CL, 0) → CL/mc, (@6E, 6E, 0) → 6E/mc, (SPY, SPY, 0) → SPY/mc, (@6E, "", 0) → 6E/parsed, (@ES, ES, 1) → ES/parsed, (@ES, ES, -1) → ES/parsed, (@ES, ES, 512) → ES/mc; SymbolExpiry 2038-01-01 / 1900-01-01 → placeholder (calendar), 2026-12-18 → used; "ES" as a stock gets no preset; empty root rejected; `RootError("", "")` is an error (F1).
MI-T3	Preset cross-check: tick/bigPointValue mismatch throws `mismatch:tick` / `mismatch:bigPointValue`. MC-internal cross-check on the four `_RootDump` rows (§4.4.3) passes: 0.01 × 100, 1E-05 × 99999.999999999985 (within 1e-9), BPV 50/1000/125000/1 equal; perturbed rows (struct PriceScale 0.1 vs 100; BPV 50 vs 5) throw `mismatch:mcPriceScale` / `mismatch:mcBigPointValue` with the message naming both values.
MI-T4	Expiry rules vs CME calendar dates per root 2016–2027 (fixture CSV with the source URL); ES rows = `EventRules.RuleExpiryDate`. **As merged (IT T4 L196–242):** `fixtures/expiry-worked.csv` (expiry analyst-checked, roll computed from the preset rule, §8.1); NyseStandIn on CL/GC/ZN/6E; 6E `BusinessDaysBeforeExpiry(5)` (L219); no weekday fallback (an Observed/None calendar refuses `calendar`, L228); `months` refusal (L241–242).
MI-T5	Roll rules: the ES set reproduces `RollDatesDigest2000To2035 = "fdd63bd8f179a8d4"` (`_RegimeGate` L60).
MI-T6	`SeriesWindow` generalised: `tests/RollTests` unchanged; CL/GC monthly windows.
MI-T7	Trading day from sessions: CME Globex template (Sun 17:00 → Mon 16:00 CT, and the ET equivalent) gives `GlobexTradingDayEt` for every 1-min close over 2016–2024 incl. DST weeks; a two-session day (GLOBEX 17:00–15:15 + 15:30–16:30, the 9-entry list of `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`) shares one key; FX 17:00 ET template = ET+7h; 00:00 UTC crypto = UTC date; a day-only template keys by calendar date; a bar outside every session throws.
MI-T8	Derived fields: `dayOpen`, `sessionEnd`, `hasOvernight`, `rthSplit` (day-only template → chart session; Globex template + ES → `EquityIndexLegacy` equal to `_CmeSessionClock` defaults field by field; other 24 h → null).
MI-T9	Cost floor in integer cents: ES 14, **MES 20**, a non-integer tickValueCents throws.
MI-T10	Sizing: caps reduce q; the off-portfolio skip logs Info; floor off by default changes nothing; q never grows.
MI-T11	Exception text per reason code (§6.3), incl. `sessions:*`. (The T11 list, IT L440, does not include `months`; `months` is tested in T4.)
MI-T12	INSTR/CFG line formats pinned; culture-invariant (run under de-DE). **As merged:** full ES INSTR line (IT L537–539) and RUN suffix format (L544) pinned (§6.1); M3 ordering, enforced/log-only and Guard tests IT L596–630.
MI-T13	`profileHash` stable and outside every `configHash`. **As merged (IT L446–476):** ES `020deca78d6392b7`, 6E `8e8b3e9b0e230e12` (L472–473), full 6E line (L475). MC (2026-11-04): real 6E chart `55576df6a8a99c05`, reproduced byte-for-byte on .NET 8 (runtime-stable). **PR-M4 (D2):** add the tz-invariance tests and the expected pins of §6.1 / §11.1 item 9. `MonthMask.Of` bad-letter test L631–640.
MI-T14	Lint (in `tests/RepoLintTests`, PR-M1d) **scoped to MC enum types** (`ESymbolCategory`, `RequestTimeZone`, `EResolution`, `BuildVolumeOn`, `MTPA_MCSymbolCurrency`, `MTPA_MCSymbolCategories`, `MTPA_SymbolFields`, `DayOfWeek` from `SessionObject`): fails on `.ToString()` of those values feeding a comparison or a parse, on `Enum.Parse`/`TryParse` with those types, and on `System.Reflection` / `GetProperty(`. Exact allow-list (file + line pattern) as of M4 @d3c0c1a: `_CustomSignal` inputSnapshot reflection, `_MarketBreadth.Strategy.CS` `using System.Reflection;`, `barZoneLabel` (log), the `SeriesKeyWarning` text (log). **`_TestPortMM` L90 removed with the file in PR-M4 (A2)** — a stale allow-list entry fails the "no stale entries" check. Repo-enum parses (`_EconomicCalendar`, `_StructuralTargetPatterns`, `_EventCalendar`, `_PortMM`) are out of scope.

ID	Test
MI-T15	ES bit-identity per PR: `_TestGate0` canary and the DVP D9a summary unchanged against the documented baseline; ES fixtures in `tests/*` unchanged; requires Ettienne's MC runs for every PR, numbers pasted into the PR. **Result 2026-11-08 (rev 2.5.2), reported native evidence at frozen `388314a`, not box-reproduced.** Ettienne 22:04 ET: "We can accept todays runs as baseline". Post-#88 exports were not found. Compile All 14:15–14:16 ET succeeded (assembly `gdbhzsw4_CS.dll`, 2,103,296 bytes; csproj SHA-256 prefix `ec02edfb` unchanged). G0 run `b04cc95a`: `_TestGate0` Canary, @ES 1-minute, 2016-11-17..2024-12-31, 35,109 trades, Strategy Performance Report net PF 0.84, $2.25 and $6.25 per side, Bar Magnifier 1 tick. Variant number not reported; not the IP-T46 set. The owner skipped the trade-list export. No SHA-256 exists for baseline `b04cc95a`; a future identity comparison needs a fresh export. D9a runs `7fbe5bff` and `86353b4f`: Data1 @ES 15-minute, Data2 @ES 60-minute, CME Globex, Trade Volume, Max bars back 50, Journal; `chartStart=1` only because `firstBarEt` is 2016-11-21 22:00; RUN-END bar 188581; 0 Error lines; `configHash` `80cf8ab6ad9ba5cd`; `profileHash` `020deca78d6392b7` both streams; Data1 matchPooled 95/95 warmEnd 2017-01-31; Data2 87/87 warmEnd 2017-10-01. Run-id-stripped DEGREE 196/196, D9A 199/199, CAL 62/62 identical. `0ea50cd4` is not the baseline. Open: MC-internal pre-execute volume-profile snapshot sync at the realtime transition (`ArgumentNullException` parameter `source`, `Enumerable.Select` inside `ChartingPlugin.Snapshot.Profile..ctor`, called from `VolumeProfilesSnapshots._full_sync` / `Syncronize` / `OnBeforeRTCalc` / `PreExecuteImpl`). No user-study frame. Single occurrence, 18:01:05.349 ET, study `_TestGate0 (@ES-1 Minute)`, PID 9640, then SwitchOff at 18:01:06. Not reproduced in the 2024-capped baseline runs. Analyst inference, not a finding: run `eee518cc`, the only run whose data reached 2026-11-06. Ettienne, verbatim: "The exception only blocks live-readiness, not the merge". Unresolved or recurring, it blocks live-readiness and is not an M5 merge gate. Check 7 unrun.
MI-T16	Session ends from the list (R14): the 9-entry GLOBEX list of `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt` with 1-min bars and `LastBarInSession` false at 15:15 (true only at 16:00, as dumped) gives session-end events at **15:15 and 16:00** each weekday and trading-day rollover after 16:00; a duplicate CalcBar at the same 16:00 time (`01M6HSDBFH2D6H21PMYXYKFJHV.txt`) resets once; a session-reset engine fixture (VPFUT/AVWAP/FLOW reset hook) sees exactly two resets per GLOBEX day and one per Default day; a list with no bar at its `EndTime` (data stops early) ends the session on the next out-of-session bar or on `LastBarInSession`.

State on `d3c0c1a` (rev 2.5): `tests/InstrumentTests/run.sh` → `InstrumentTests: 756 passed, 0 failed` on the M4 head (box, .NET 8, 2026-11-04 ET); full check **23 suites, 4528 passed, 0 failed**. Pins: 6E `55576df6a8a99c05`, ES `6cff053d1b215324`, AAPL `41886799d411dd0d`.

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11. PR list (each small; each targets `main`, opened after the previous one merged; no stacked PRs)

Status on `d3c0c1a` (rev 2.5): M0 **#89** merged, M1a **#90** merged, M1b **#91** merged, M1c **#92** merged, M1e **#93** merged, M1d **#95** merged, M2 **#96** merged, M3 **#97** merged, **M4 #98 merged** (`d3c0c1a`, final head `2418052`).
Extra, outside this list: #94 `_ExitHalfRiskHalfRoll` exit-only (half off at +1R, trail the rest; exit-only invariant; `OrderExit.Total`), merged (`ed3e4a5`); it changes the legacy study only. Next: **PR-M5**.

#	PR	Content	Behaviour change	Tests
—	**#88**	Land #86 + #87 (existing)	—	—
0	PR-M0 **#89 merged**	`_InstrDump.Strategy.CS` (§4.4, derives from `_CustomSignal`) + the `LogDirectory()` accessor in `_CustomSignal` (§4.4.1) + csproj entry + import note; `MCInstrDump_*.txt` output committed as fixtures after Ettienne's runs	none	existing suites
1	PR-M1a **#90 merged**	F1: `_MtfMath.SymbolRoot` (§4.3 steps 1–2: leading digits + letter run, month drop, "/" removed; L109–119 on `6853398` before #90, L120–135 after #90), doc L28, `RootError` empty = error, roll inheritance and same-root checks	digit-led roots / FX only	MI-T2 (subset) in existing MtfMath/Dvp tests
2	PR-M1b **#91 merged**	F2 (`isFuturesCategory` at L1069/L1866/L1882), F4 (tick throw), L2462 typed	FutureRolover, bad tick	existing suites
3	PR-M1c **#92 merged**	F3a + F3b in `_ExitHalfRiskHalfRoll` only	legacy study only	—
4	PR-M1e **#93 merged**	Retire `_InstrProbe` + `_InstrProbeCore` + `tests/InstrProbeTests` + csproj L521–522 + sln entry	none	suite count 23 → 22
5	PR-M1d **#95 merged**	MI-T14 lint + allow-list	none	MI-T14
6	PR-M2 **#96 merged**	`_InstrumentProfile.Function.CS` core (enums, resolver, presets, `InstrumentConfig`, `CostModel`, exceptions, session derivations on structs), `<Compile Include>` + import order, `tests/InstrumentTests` + sln, rev-3 "superseded for C1–C9" banner (Q10)	none (not wired)	MI-T2–T9, T11, T13, T16 (core)
7	PR-M3 **#97 merged**	Wire `buildProfiles()` + `SessionReader` (StartCalc read + first-CalcBar re-check), `ClassOf` switch, INSTR line, cross-checks, `profileHash` on RUN	log only (+ `mismatch:*` and `sessions:changed` throw, §6.2)	MI-T12, T13, T15 (MC run pending)
8	PR-M4 **#98 merged** (`d3c0c1a`, head `2418052`)	Enforce every §6.3 refusal globally (A1); D1 class:root cost keys; D2 exchange-time profileHash via `exchangeZoneIdOf` token match; strict tick / FALLBACK_TICK removed; floor `src=…`; PORT publish line deferred to first CalcBar; `_TestPortMM` deleted (A2); DST tests; Start marks started only after Guard(Build). Scope detail: §11.1	refusals → Error + throw on every stream for all `_CustomSignal` studies; same instrument hashes equal across chart/PT zones	MI-T7, T8, T15, T16 + D2/DST; full check **23 / 4528 / 0**
9	PR-M5	Expiry/roll from the profile: `_RollDetector`, `EventRules`, host quarterly loops (`_DvpEngineTest`, `_Test6`). Rev 2.5.1: separate `ownContract` / `sharedCalendar`; no roll facts → `Unavailable` on Step/Classify (not the ES quarter, not false); explicit queries throw S5	non-ES only; ES identical	MI-T4–T6, T15
10	PR-M6a	G0: tick default removed + **Gate0Tests 35 constructions set tick**; root; S3	none for ES	Gate0Tests, T15
11	PR-M6b	FLOW: chunk boundary = listed `sessionEnd` converted to ET. Stale Globex ES moves 17:00 → 17:30 ET	ES on that template moves 30 minutes (accepted 2026-11-09)	OrderFlowTests
12	PR-M6c	DAILY/REGIME `sessionsPerYear` (defaults only; 63-session window stays)	non-futures only	DailyTests, RegimeTests
13	PR-M6d	LEVEL/AVWAP tick defaults removed (hosts already set `tickSize()`)	none	AvwapTests, LevelStudyTests

#	PR	Content	Behaviour change	Tests
14	PR-M6e	RANGE/WEEKLY/PSTATS: S7 for `RthSessions`; CFG note; ringBars literal	none for ES	InstrumentTests
15	PR-M6f	PORT cluster from profile	none	PortMM tests
16	PR-M6g	DVP roll table from profile	none for ES	DvpTests
17	PR-M6h	REGGATE: ES only (N2), explicit hash switches, S3	hash identical	RegimeGate tests, T15
18	PR-M7	Cost floor (opt-in, new) + costs in cents	none unless enabled	MI-T9, T10
19	PR-M8	Calendars for other roots, only when sourced (ES table unchanged)	non-ES only	calendar fixtures
later	PR-M9	Self-calibrating roll gap (Q4); ES stays 8 until it reproduces `tests/RollTests` and the m-event splices; `tools/m-event/out/roll_splices_2016_2024.csv` is read-only	none for ES	RollTests

11.1 PR-M4 scope — DONE (rev 2.5; merged #98 @d3c0c1a)

1. **Log-only → enforced (A1, Ettienne 2026-11-04):** "Keep the strict version: stop any strategy on any bad stream." Every refusal of §6.3 (incl. `class`, `unknown-field:*`, `tick`, `currency`, `calendar`, `months`, `sessions:*`, `tz:bar`, `tz:exchange`, `mismatch:*`) is Error + throw **globally at StartCalc on every data stream for all `_CustomSignal`-based studies**, not only for engines that use the field. The `INSTR ds(n) log-only…` line and the null-profile / `profileHash(…=none)` path are gone. `InstrumentStreams.Start` sets `m_started` / `m_pending` only after `Guard(Build)` returns. **Refused:** Unsupported/unmapped class; non-USD; unresolved bar/exchange zone (unmapped exchange → `tz:exchange`); PriceScale ≤ 0 / tick ≤ 0; missing or changed sessions; failed typed/session reads; invalid `InstrumentConfig` (incl. old root-only cost keys → `unknown-field:config`); `mismatch:*`; `calendar` / `months` when rules need them. **Still runs:** futures with no preset (cost/floor Unknown); USD stocks on NYSE/NASDAQ/ARCA.
2. **ClassOf:** an unmapped `ESymbolCategory` in `buildProfiles()` is Error + throw (same as `classOfData1`). **DONE.**
3. **FALLBACK_TICK audit (R6):** `FALLBACK_TICK = 0.01` removed from `_VolumeProfileBase.Strategy.CS` and `_VpConfluenceEngine.Function.CS`; tick comes from the Data1 profile (or the strict host tick when M-INSTR is off). **DONE.**
4. **`tickFromMinMove` audit:** the `PriceScale == 0 → MinMove` path and `Math.Abs` of a negative ratio are gone; a tick that cannot be resolved is Error + throw. Host `tickSize()` prefers the Data1 profile tick. **DONE.**
5. **Floor-Unknown reason** on INSTR: `floor(Unknown src=no-preset|no-cost|cost-class-mismatch|config-invalid)`. **DONE.**
6. **Portfolio Trader test, before M4 starts: DONE 2026-11-04** (Ettienne, `5c9ccd2`): one PT log for @6E, @ES, AAPL (§4.4.2).
7. **`eMTPA_SF_Root`: DONE 2026-11-04**: MI-T1 compile passed, `rootSrc=mc` on 6E/ES/AAPL (§4.1); the MC 6E hash `55576df6a8a99c05` reproduces on .NET 8 (`8e8b3e9b0e230e12` is the synthetic-list pin).
8. **Cost keying by asset class + root (D1): DONE.** `CostKey(EAssetClass, root)` = `<Class>:<ROOT>`; defaults `Future:ES` 1700, `Future:MES` 250; `Validate` rejects root-only and unknown-class keys (`unknown-field:config`); a same-root row of another class is not used (`cost(Unknown)`, `floor(Unknown src=cost-class-mismatch)`, one Warning).
9. **profileHash in exchange time (D2): DONE.** `Canon` hashes sessions and rth in the exchange zone via `exchangeZoneIdOf` (token match, §6.1); refusals `tz:bar` / `tz:exchange`; INSTR adds `xsessions(…)`; engine-facing fields stay in bar time. Pins confirmed in IT: 6E Default `55576df6a8a99c05`, ES Default `6cff053d1b215324`, AAPL `41886799d411dd0d`. DST tests (summer, spring-forward, fall-back) keep the 6E pin. **Still OPEN:** GMT chart VERIFY + live check of composite exchange strings (**A3**, §12.4).
10. **PORT publish line timestamp: DONE.** `PORT publish on: …` deferred from StartCalc to the first CalcBar (calc thread: bar date + zone known).
11. **`_TestPortMM` deleted (A2, Ettienne 2026-11-04).** File `git rm`'d; MI-T14 allow-list entry removed; `_TestPortArm` remains (header notes the deletion). **Open:** live-check-sheet Check 7 MM-side checks need a replacement MM test signal before that check can run (§12.4).

Every PR lists the R1–R14 rules it touches, the VERIFY items it closes, its csproj/sln changes and the PowerLanguage Editor imports, and re-reads its line cites on current main.

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12. Questions

12.1 DECIDED

- **Q1 / Q2 Sessions and RTH windows: DECIDED by Ettienne 2026-11-02.** Verbatim: "The trading hours in questions 1 and 2 should be driven from Multicharts - it has session information you can get and use. This info is setup from the chart and just work with that. Each instrument also has a 'Default' setting which should be correct and used the traders in general." Applied as R12 and §4.5. Q2 (RTH windows for CL/GC/ZN/6E) is moot: no RTH windows in presets.
- **Q3 Costs for other roots: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): IBKR fees + one tick slippage, as ES/MES. **Data input outstanding:** IBKR per-side fees for NQ, MNQ, CL, GC, ZN, 6E; until given those roots have cost Unknown.
- **Q4 Roll-gap threshold: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): self-calibrating in PR-M9, only after it reproduces RollTests and the m-event splices; ES keeps 8 until then.
- **Q5 Roll rules for CL/GC/ZN/6E: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): rule-based table per §8.1, confirmed against the vendor's continuous splice; series detection fallback.
- **Q6 Retire `_InstrProbe`: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): PR-M1e.
- **Q7 Data-availability notes: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): `dataNotes`, logged on INSTR; FLOW bid/ask features undefined before the date (approved degraded mode D6, §6.4). **Data input outstanding:** dxFeed tick bid/ask start dates for @ES and @MES.
- **Q8 Non-USD: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): USD only; non-USD throws `currency`.
- **Q9 Stop floor: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): cost-derived, integer cents (§9.4), opt-in (PR-M7).
- **Q10 Supersede rev 3: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): banner in PR-M2.

12.2 N1–N6: DECIDED (Ettienne 2026-11-03 09:47 ET: "1 to 5 is fine, on 6 keep that dump in the same directory as the created standard log file")

- **N1 RTH split on a 24-hour chart: DECIDED as recommended (B).** ES/MES/NQ/MNQ on a 24-hour chart keep today's equity-index split in code (`rth src=EquityIndexLegacy`, ES bit-identical); other roots on a 24-hour chart throw S7 for RTH-dependent engines; on a day-only chart the chart session is the RTH. Option A (a second RTH-template stream) is not used because of MC-1439 stalls under Realtime-History Matching.

- **N2 A2-63 on MES: DECIDED as recommended.** REGGATE accepts ES only; MES would be a new trial with its own pre-registration.
- **N3 Overnight-only outputs without an overnight segment: DECIDED as recommended.** NaN/false with one Info line (U1–U2), not a throw.
- **N4 TOD half-day flag without a verified calendar: DECIDED as recommended.** Approved degraded mode D7 (§6.4).
- **N5 Asia/Europe/NY clock windows: DECIDED as recommended.** Kept as instrument-independent code defaults in ET (VPFUT L52–63, `_CmeSessionClock` L48–61), clipped to the chart's sessions; not presets.
- **N6 `_InstrDump` output: DECIDED.** Same directory as the standard MCSignal log, via the host's resolver (`_CustomSignal LogFileSingleton` L71–91, exposed through a new `LogDirectory()` accessor); file `MCInstrDump_<symbol>_<yyyyMMdd-HHmmss>-<pid>.txt` (§4.4.1).

12.3 Data inputs still pending (do not block PR-M0 to PR-M3)

- Q3: IBKR per-side fees for NQ, MNQ, CL, GC, ZN, 6E (until given: cost Unknown, S5).
- Q7: dxFeed tick bid/ask start dates for @ES and @MES.
- PR-M0 OPEN items (§4.4.2): runs not yet made: @ZN, one FX pair, one crypto symbol. The Portfolio Trader per-symbol item is **CLOSED** (PT test passed 2026-11-04, §4.4.2, §11.1 item 6).
- **MI-T1 / MI-T15 (Ettienne): DONE 2026-11-04 on `5c9ccd2`.** Compile All passed (one-time import of `_InstrumentProfile.Function.CS`; `eMTPA_SF_Root`, `eMTPA_MCSC_USD` compile). 6E chart (Exchange tz, Default, BVO TickCount): `tick(5E-05) bpv(125000) tickValue(6.25)`, NyseStandIn note, `roll(BusinessDaysBeforeExpiry(5))`, profileHash `55576df6a8a99c05`, reproduced byte-for-byte on .NET 8; the pins `020deca78d6392b7` / `8e8b3e9b0e230e12` are for the 9-session GLOBEX list. The 2026-11-04 text said they were not expected in MC. **Corrected in rev 2.5.2 (§6.1):** `020deca78d6392b7` is the ES canon of that list, and the Nov 8 CME Globex D9a chart printed it. `8e8b3e9b0e230e12` remains the 6E row of the same list and was not printed by that ES run.

12.4 Decisions (current through rev 2.5.2)

- **D1 Cost keying: DECIDED by Ettienne, 2026-11-04: key costs by asset class + root** (e.g. `Future:ES`), so a configured futures cost never applies to a stock with the same root. Rule, key format, default mapping (`Future:ES` 1700, `Future:MES` 250), class-mismatch handling and the configHash check: §5.3; floor reasons: §9 4a; implementation: **DONE in PR-M4** (§11.1 item 8). M3 (`5c9ccd2`) keyed by root only; M4 keys by `CostKey(class, root)`. Options considered: (a) keep root-only plus the preset gate; (b) class + root (recommended, chosen).
- **D2 profileHash time zone: DECIDED by Ettienne, 2026-11-04: compute profileHash from the sessions expressed in exchange time**, so the same instrument hashes the same whatever the chart / Portfolio Trader time zone. Evidence: @6E `55576df6a8a99c05` (chart, Exchange tz, Sun 17:00→Mon 16:00 … Central) vs `9523b25025f9ed5b` (PT, Local, 18:00→17:00 ET); both reproduced on the box from the canon. Zone source, refusals (`tz:bar`, `tz:exchange`), conversion, INSTR `xsessions(…)`, rth handling and expected pins: §6.1; implementation and tests: **DONE in PR-M4** (§11.1 item 9). M3 (`5c9ccd2`) hashed bar-time sessions; M4 hashes exchange-time sessions via `exchangeZoneIdOf`.
- **A1 Enforcement scope: DECIDED by Ettienne, 2026-11-04 (own words):** "Keep the strict version: stop any strategy on any bad stream." M4 enforces every §6.3 refusal globally at StartCalc on every data stream for all `_CustomSignal`-based studies — not only for engines that use the field. Refused vs still-runs list: §11.1 item 1 / §6.2.
- **A2 `_TestPortMM`: DECIDED by Ettienne, 2026-11-04 — delete it.** `_TestPortMM.Signal.CS` removed in PR-M4; its MI-T14 allow-list entry removed; `_TestPortArm` remains (comments updated). **Open item:** live-check-sheet Check 7 MM-side checks (P5b L1–L9) need a replacement MM test signal before that check can run (`docs/engines-manual.md` notes the removal).
- **A3 Composite exchange names + GMT VERIFY: OPEN.** `exchangeZoneIdOf` requires exact tokens; composite / feed names (NYSEARCA, NASDAQGS, CMEGLOBEX, AMEX, BATS, CFE, and similar) are deliberately unmapped → `tz:exchange` until a live check of real `Bars.Info.Exchange` strings on dxFeed shows what the feed reports. Separately, one MC run on a **GMT** chart is still VERIFY (does MC shift GMT sessions across DST? the box conversion uses the first bar's week).
- **MI-T15 baseline: DECIDED by Ettienne, 2026-11-08 22:04 ET (own words):** "We can accept todays runs as baseline". The Nov 8 reported native runs replace the missing post-#88 G0/D9a exports as the documented baseline (revision log, MI-T15). Evidence only. **Exception vs merge: DECIDED by Ettienne, 2026-11-08 about 22:06 ET (answer t515, own words):** "The exception only blocks live-readiness, not the merge". The 18:01 `ArgumentNullException` blocks live-readiness if it recurs or stays unresolved. It is not an M5 merge gate. **Trade-list hash: skipped.** The owner did not choose a G0 export. No SHA-256 exists for baseline `b04cc95a`. A future identity comparison needs a fresh export. Check 7 is not part of either decision.
- **Own-contract controls: DECIDED by Ettienne, 2026-11-06 12:13 ET (own words):** "I approve separate expiry/roll controls in M5, preserving other calendar events, explicitly marking unsupported contract-expiry status unavailable, and rejecting explicit unsupported expiry/roll queries." Applied in PR-M5 (§6.4 own-contract availability, §7 EVENT, rev 2.5.1). Shared calendar rows stay independently selectable. No new `[Input]`, no per-asset-class event mask, no NyseStandIn session count, no cost-arithmetic change.
- **Accepted, recorded:** `MonthMask.Of` throws `ArgumentException` on a bad month letter **without** an Error line (exception to "Error before every throw"). It is called only on static preset data, the doc comment says so, and IT L631–640 tests it; a bad letter fails every test run before any MC run. No change.

12.5 Code vs spec divergences found for rev 2.4 (on `5c9ccd2`; several closed by PR-M4 @d3c0c1a — see V6, V7, V8, V15 below)

#	Where	Spec said (rev 2.3)	Code does	Rev 2.4
V1	§5.1 `calendar`	`{CmeEquityIndex, Nyse, Observed, None}`	adds `NyseStandIn` (IP L33)	spec updated
V2	§8.1 CL/GC/ZN/6E	calendar Observed; 6E roll "Mon of the expiry week"	NyseStandIn; 6E `BusinessDaysBeforeExpiry(5)`	spec updated (Ettienne's M2 decisions)
V3	§4.3 step 8	6E tick `5.0000000000000007e-5`	raw `5.0000000000000009E-05` (R, Framework); profile `5E-05`	spec corrected
V4	§5.1 `dataNotes`	list of (from, what)	one string per root	spec updated
V5	§6.1 example	no `rootSrc`, no `dataNotes`, `sessionEnd(16:00)`, n=1	`root(ES rootSrc=mc)`, optional `dataNotes(…)`; GLOBEX 9-entry list ends 16:30	pinned lines added
V6	R13/R6 "Error before every throw"	every refusal Error + throw	M3 log-only refusals wrote only a Warning; enforced ones wrote the Error	**CLOSED by M4 (A1):** every refusal Error + throw
V7	R6	no silent fallbacks		**CLOSED by M4** (§11.1 items 3–4)

#	Where	Spec said (rev 2.3)	Code does	Rev 2.4
			`FALLBACK_TICK` in VP/VpConfluence; `tickFromMinMove` PriceScale==0 → MinMove and `Math.Abs`	
V8	§4.2	unmapped category throws	`buildProfiles` turned it into a log-only Warning (`classOfData1` still threw)	**CLOSED by M4** (§11.1 item 2)
V9	§5.1/§9 floor	floor from cost	only with a matched preset and a valid config; cost still shown without a preset	documented; floor reason M4
V10	R6 `calendar`	rules without a calendar throw	Build refusal is log-only in M3; reachable only via an override row (presets all have a calendar; `Validate` does not reject it)	documented; M4
V11	§6.2	"null profile" on refusal	a first-bar session refusal keeps the StartCalc profile (INSTR still printed, `sessions(n=0)` when none were read)	documented
V12	MI-T11	every reason code	T11 list omits `months` (covered in T4)	noted
V13	§4.1	`eMTPA_SF_Root = 1` stated as fact	mirror-only; not in the PG or a dump	closed 2026-11-04: member compiles (MI-T1); value not relied on
V14	§5.3 costs	class + root keys (D1, 2026-11-04)	root-only keys (`ES`, `MES`); a stock `ES` shows the future's cost	PR-M4 (§11.1 item 8)
V15	§6.1 profileHash	sessions/rth hashed in exchange time (D2)	M3 hashed in bar time; @6E `55576df6a8a99c05` (Exchange) vs `9523b25025f9ed5b` (PT Local)	**CLOSED by M4 (D2):** exchange-time hash via `exchangeZoneIdOf`; pins 6E `55576df6a8a99c05`, ES `6cff053d1b215324`, AAPL `41886799d411dd0d`

Appendix A. Review responses (Coder review of rev 1, 37 items; each re-checked on `7dbdb1b`)

#	Item	Response	Where
1	Lint fails on main; needs allow-list, MC enums only	Accepted. Scoped to MC enums with an exact allow-list; remaining MC-enum string uses listed as findings	§3.3, MI-T14, PR-M1d
2	Removing `_InstrProbe` breaks InstrProbeTests	Accepted. Own PR with the suite, csproj L521–522 and sln; M1 does not depend on the M0 dump	PR-M1e
3	Removing G0 tick default breaks Gate0Tests	Accepted, with a correction: 35 `new Gate0Engine` constructions and **none** sets the engine tick (L168/184/191 are constructions); all 35 updated in PR-M6a	§7, PR-M6a
4	ringBars not ES-unchanged if derived	Accepted. Literals kept (2880; `_ProfileStats` 1440); they are bar counts	rows 55, 65; §7
5	§9.1 vs willArm	Accepted. §9 restates the code; floor is new opt-in	§9
6	Throw vs degrade conflicts	Accepted. One rule (R13), every case listed; rev-1 degrades withdrawn	R13, §6.4
7	Row 13 cite	Accepted: L1840–1842	row 13
8	Row 23 missing `MultiChartsProfile=0`	Accepted	row 23
9	Row 24 value window (09:30,16:00]	Accepted	row 24
10	Rows 26/28 VPFUT own table	Accepted	rows 26, 28
11	Counts	Accepted. One area per row, totals 70	§3.1
12	L2621 NYMEX/COMEX → Eastern	Accepted as a finding; map kept only for ET conversion; fixed in M4 if dump A3 shows Central	row 4, §4.4 A3, §4.5
13	Host quarterly tables	Accepted	row 64, PR-M5
14	L1888 Eastern	Accepted; removed by session-derived day key	row 63, §4.5
15	`_ProfileStats` 4-tick; NQ $2.50	Accepted	rows 66, 67
16	`SymbolRoot` doc	Accepted	row 68, PR-M1a
17	F1 wider	Accepted; MI-T2 adds 6E/6B and EUR/USD vs EUR/JPY	§3.2, MI-T2
18	F2 also L1069/L1866	Accepted	§3.2, PR-M1b
19	F3 description + second bug	Accepted	rows 60, 70; PR-M1c
20	F4	Accepted	PR-M1b
21	`Bars.Sessions` compiles	Accepted; StartCalc population, zone and per-series sessions CONFIRMED by the PR-M0 dump (rev 2.2); read at StartCalc, re-checked on the first bar	§4.1, §4.4 A2–A4, §4.5
22	`StrategyCurrencyCode` on plain signal	Accepted; moved to section B	§4.4 B2
23	Drop Description/PointValue/Resolution.Size from dump	Accepted	§4.4 A1
24	Dump3 hygiene	Accepted; cited as box-only run output; M0 commits typed output	header, §4.4
25	Rename to `bigPointValue`	Accepted	§5.1
26	Split M1	Accepted (a, b, c, d + e for the probe)	§11

#	Item	Response	Where
27	Sequential PRs on main	Accepted; #88 first, no stacking	header, §11
28	M2 build plumbing	Accepted	PR-M2, §10 header
29	MI-T1 cannot be a test	Accepted; host exhaustive switch	§4.2, MI-T1
30	MI-T15 baseline + MC runs	Accepted; post-#88 baseline, Ettienne's runs per PR	MI-T15
31	Banner in M2	Accepted	PR-M2
32	R3 PriceAction preset	Accepted; preset clause dropped, "PriceAction unchanged" kept	R3
33	R4 wording	Accepted: "no new [Input]s"	R4
34	REGGATE Q8 cross-ref; hash switch	Accepted; cross-ref fixed, MES as new trial raised as N2, explicit switches	§7, N2, PR-M6h
35	Cost floor exact integers	Accepted; integer cents, MES = 20 test	§9.4, MI-T9
36	FLOW boundary for FX	Changed: boundary = the chart's session end (no per-class rule); spot FX is already Unsupported for FLOW (rev-3 reason 11); 6E/FX futures use their own session end	§7, PR-M6b
37	Q4: ES stays 8; CSV read-only	Accepted	Q4, PR-M9

Rejected outright: none. **Partly changed:** item 3 (no Gate0 test sets the tick, so all 35 are updated) and item 36 (session end instead of a per-class rule).